



PAN3029 / PAN3060 系列

产品说明书

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低功耗远距离无线收发芯片

PAN3029 / PAN3060 is a low-power long-distance wireless transceiver chip using ChirpIoT™ modulation and demodulation technology. It supports half-duplex wireless communication and operates in the frequency bands of 137~187MHz, 205~281MHz, 408~550MHz and 816~1110MHz. The chip has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long transmission distance. It has a maximum sensitivity of -143dBm and a maximum output power of 20dBm, which produces an industry-leading link budget, making it the best choice for long-distance transmission and applications with extremely high reliability requirements.

Key Features

- Operating frequency bands:
137~187MHz, 205~281MHz, 408~550MHz,
816~1110MHz
- Modulation mode: ChirpIoT™
- Transmit output power: -30dBm~20dBm
- Maximum link budget: 163dB
- Sensitivity as low as: -143dBm@62.5kHz
- Operating current
 - Deep sleep current: 260nA
 - Receive current: 4.1mA@DCDC mode
 - Transmit current:
 - 96mA@20dBm
 - 13mA@-2dBm
- Supported bandwidth
 - PAN3029: 62.5kHz、125kHz、250kHz、500kHz
 - PAN3060: 125kHz、250kHz、500kHz
- SF factor support, automatic SF identification
 - PAN3029: 5~12
 - PAN3060: 5~9
- Support code rate: 4/5, 4/6, 4/7, 4/8
- Support CAD function
- Support low rate mode: 0.08~59.9kbps

- Support 4-wire SPI / 3-wire SPI / I2C configuration interface, support 4 GPIOs
- Operating voltage:
1.8V ~3.6V (DCDC mode 2V~3.6V)
- Operating temperature: -40°C~85°C
- Package: QFN28 (4×4mm)

Typical Applications

- Smart factory
- Smart water
- Smart agriculture
- Smart healthcare
- Smart community
- Smart fire protection

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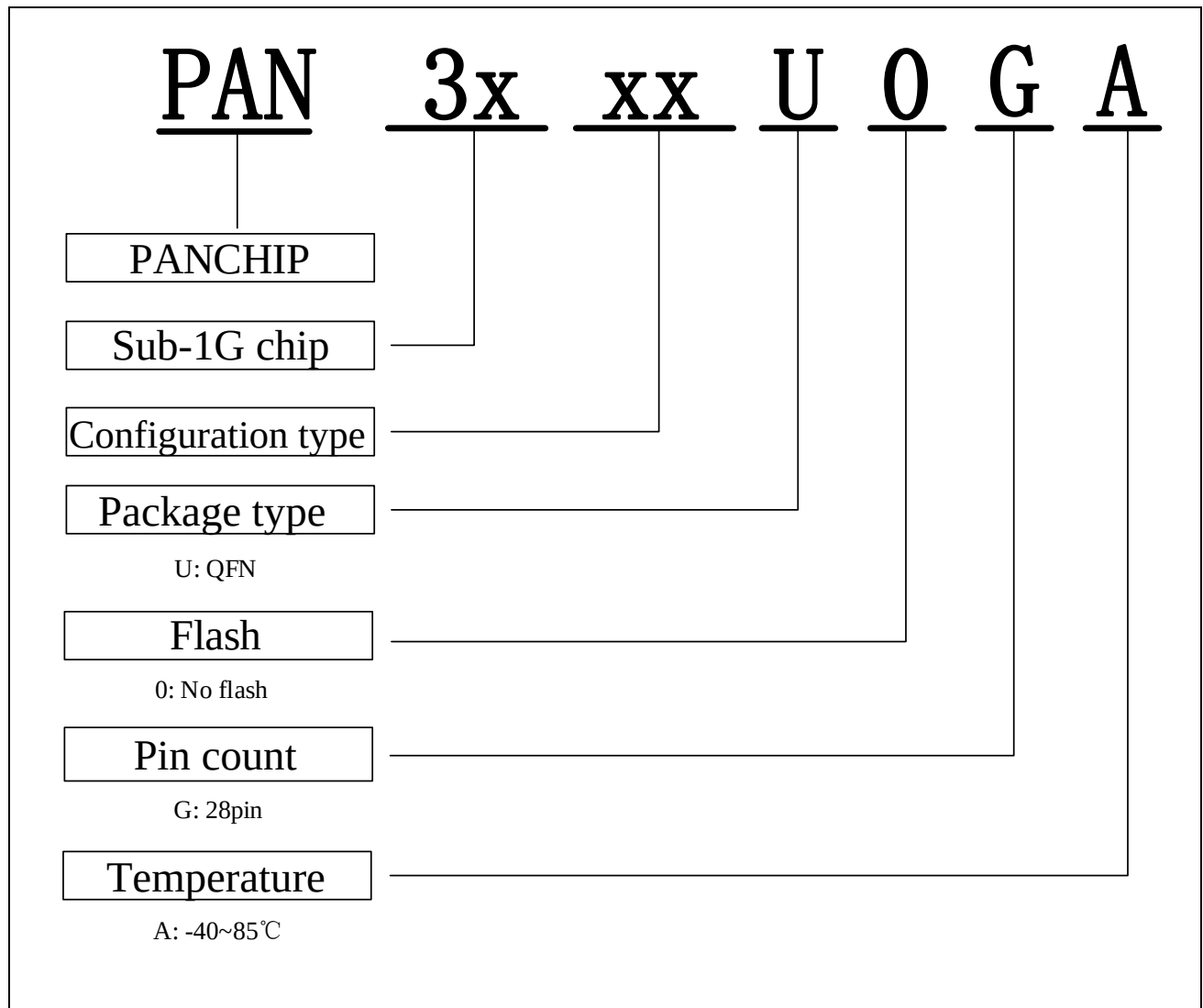
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1 Naming convention



2 Ordering Information

Product Model	Chip Type	Package	Pin count	IO	Bandwidth	SF	Bit Rate	Temperature	Packaging
PAN3029 U0GA	Sub-1G	QFN	28	4	62.5kHz 125kHz 250kHz 500kHz	SF5~SF12	0.08~ 59.9kbps	-40~85°C	Tape & Reel
PAN3060 U0GA	Sub-1G	QFN	28	4	125kHz 250kHz 500kHz	SF5~SF9	0.5~ 59.9kbps	-40~85°C	Tape & Reel

Before ordering, please consult sales for the latest mass production information.

This manual describes the models based on the largest package and most complete functions. For configuration differences between models, please refer to the ordering information on this page.

3 System structure block diagram

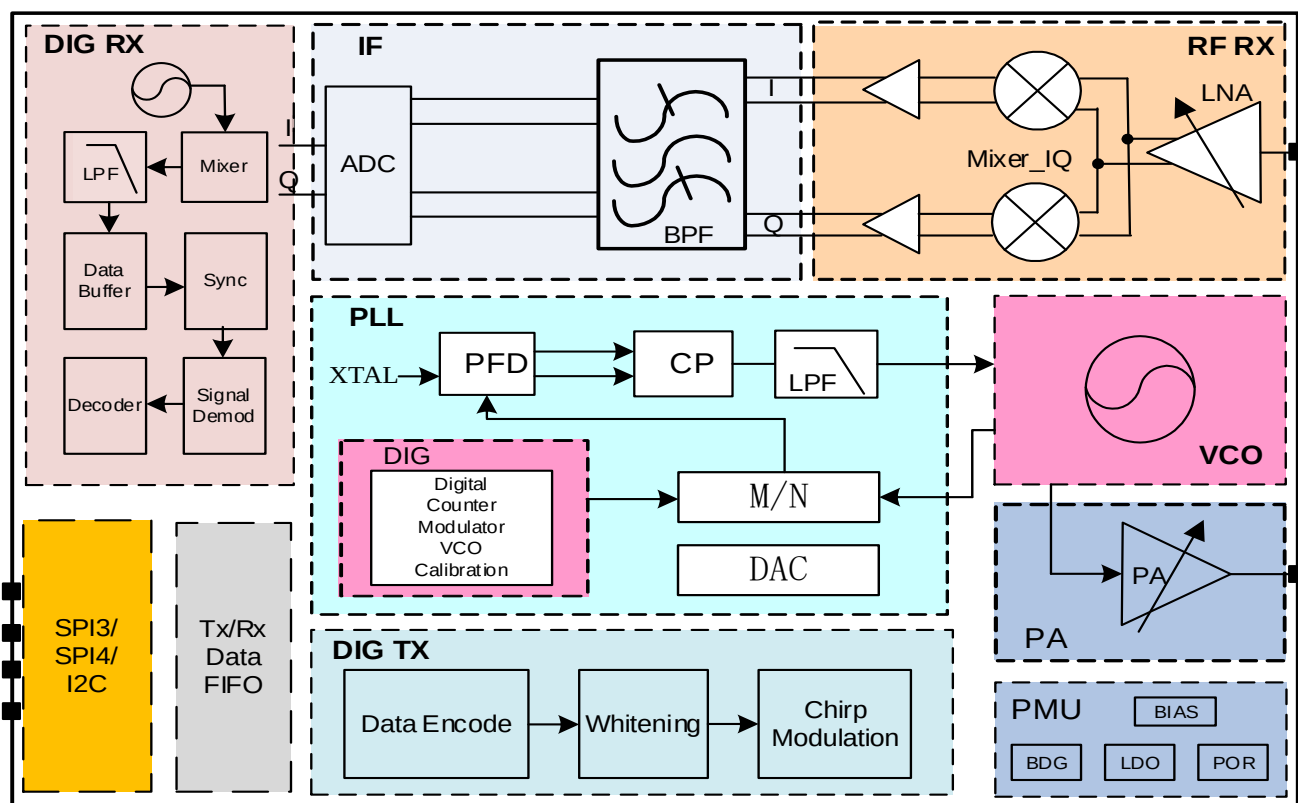


Figure 3-1 System Block Diagram

4 Pin Definition and Description

4.1 Pin Diagram

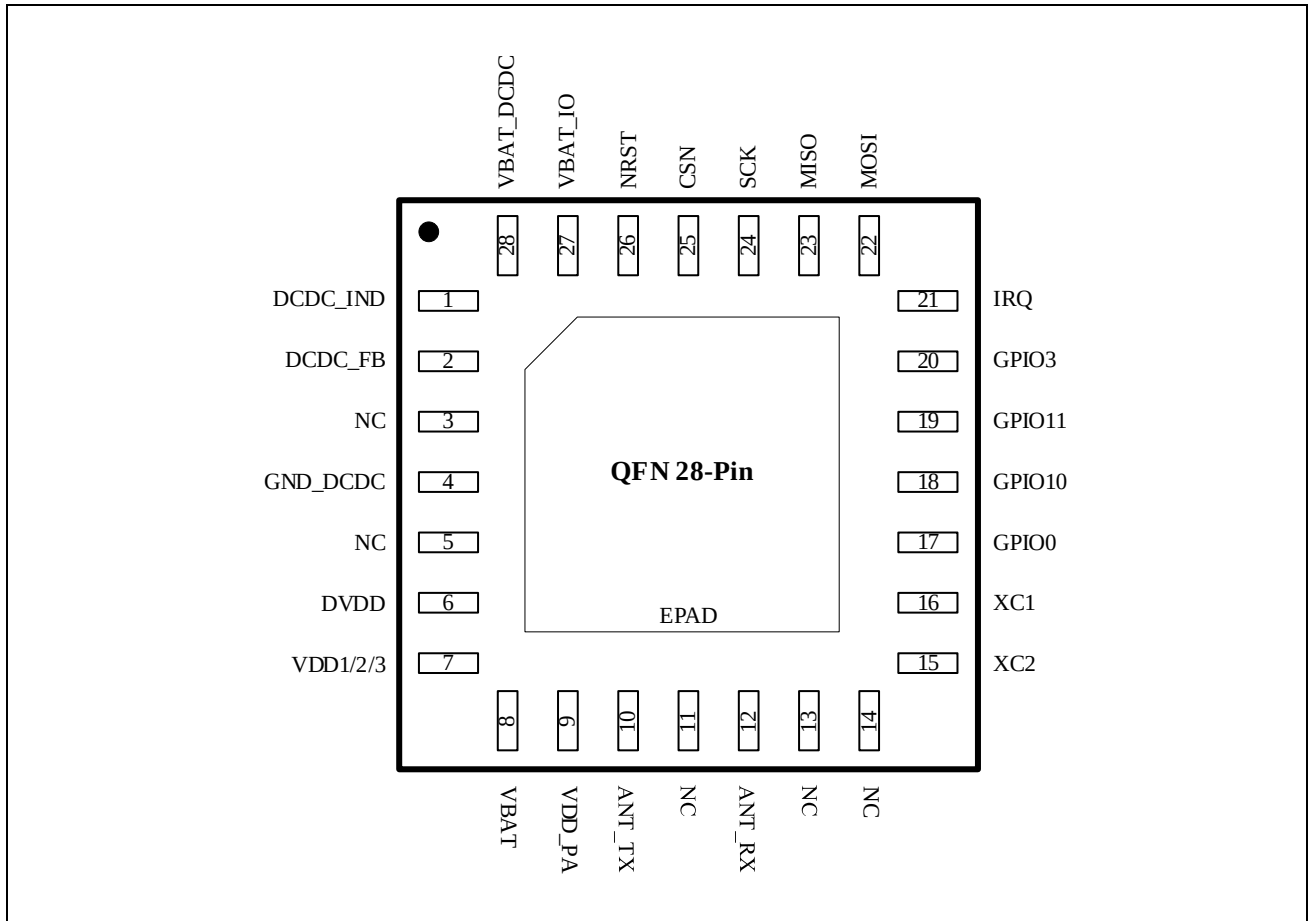


Figure 4-1 Pin Assignment

4.2 Pin Description

Table 4-1 Pin Description

No.	Symbol	Type	Function
1	DCDC_IND	AO	Internal DCDC output, connects to an external series inductor
2	DCDC_FB	AI	Internal DCDC feedback input, connects to VDD1/2/3 in DCDC mode
3	NC		For optimal performance, connect to GND_DCDC
4	GND_DCDC	G	Analog ground
5	NC		For optimal performance, connect to GND_DCDC
6	DVDD	P	Digital power, LDO output
7	VDD1/2/3	P	Analog power, connects to VFB in DCDC mode, connects to the main power supply in non-DCDC mode
8	VBAT	P	Analog power, connects to the main power supply
9	VDD_PA	P	PA power LDO output
10	ANT_TX	AO	Transmitter PA output
11	NC		For optimal performance, connect to GND
12	ANT_RX	AI	Receiver LNA input
13	NC		For optimal performance, connect to GND
14	NC		For optimal performance, connect to GND
15	XC2	AI	Crystal Oscillator Input
16	XC1	AO	Crystal Oscillator Output
17	GPIO0	I/O	Digital IO
18	GPIO10	I/O	Digital IO
		O	External PA control signal
19	GPIO11	I/O	Digital IO
		O	Channel status indicator signal
20	GPIO3	I/O	Digital IO
21	IRQ	O	Interrupt signal
22	MOSI	I	SPI data input signal
23	MISO	O	SPI data output signal
24	SCK	I	SPI serial clock
25	CSN	I	SPI chip select signal
26	NRST	I	Reset
27	VBAT_IO	P	Digital GPIO power supply, connected to the main power supply
28	VBAT_DCDC	P	DCDC power supply, connected to the main power supply



PAN3029 / PAN3060 series wireless transceiver chips

29	E-PAD	G	GND pad on the bottom of the chip, needs to be grounded
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5 Electrical characteristics

All parameters in this section are based on testing at $T_A=25^{\circ}\text{C}$ and a 3.3V supply voltage, unless otherwise specified.

5.1 Absolute Maximum Ratings

Table 5-1 Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
VDD	VDD1/VDD2/VDD3/VBAT/VBAT_IO	Frequency: 490MHz	-0.3	3.3	3.6	V
V _I	Input Voltage		-0.3	-	VDD	V
V _O	Output Voltage		VSS	-	VDD	V
T _{OP}	Operating Temperature		-40	-	85	°C
T _{STG}	Storage Temperature		-55	-	125	°C

Caution:

Exceeding one or more of the limiting values may cause permanent damage to the chip.

This device is electrostatically sensitive. Please comply with protective regulations when handling.

5.2 DC characteristics

Test conditions:

Frequency: 490MHz

Table 5-2 Voltage and current (frequency 490MHz)

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VDD	Power Supply	1.8	3.3	3.6	V	$T_A=25^{\circ}\text{C}$, LDO mode
		2	3.3	3.6	V	$T_A=25^{\circ}\text{C}$, DCDC mode
VSS	Ground	-	0	-	V	-
I _{DeepSleep}	Deep Sleep Current	-	260	-	nA	-
I _{Sleep}	Sleep Current	-	620	-	nA	-
I _{TX,20dBm}	TX Mode Operating Current	-	96	-	mA	20dBm Output Power
I _{TX,18dBm}	TX Mode Operating Current	-	77	-	mA	18dBm Output Power
I _{TX,-2dBm}	TX Mode Operating Current	-	13	-	mA	-2dBm Output Power
I _{RX,LDO}	RX Mode Operating Current	-	7.8	-	mA	LDO Mode, maximum LNA gain
I _{RX,DCDC}	RX Mode Operating Current	-	4.1	-	mA	DCDC mode, maximum LNA gain
V _{OH}	Output High Voltage	VDD-0.3	-	VDD	V	-
V _{OL}	Output Low Voltage	VSS	-	VSS+0.3	V	-

V _{IH}	Input high level voltage	0.8*VDD	-	-	V	-
V _{IL}	Input low level voltage	-	-	0.2*VDD	V	-
SPI_rate	SPI rate	-	-	10	Mbps	-

Test conditions:

Frequency: 915MHz

Table 5-3 Voltage and current (frequency 915MHz)

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VDD	Power Supply	1.8	3.3	3.6	V	T _A = 25°C, LDO mode
		2	3.3	3.6	V	T _A = 25°C, DCDC mode
VSS	Ground	-	0	-	V	-
I _{DeepSleep}	Deep Sleep Current	-	260	-	nA	-
I _{Sleep}	Sleep Current	-	620	-	nA	-
I _{TX,20dBm}	TX Mode Operating Current	-	100	-	mA	20dBm Output Power
I _{TX,18dBm}	TX Mode Operating Current	-	81	-	mA	18dBm Output Power
I _{TX,-2dBm}	TX Mode Operating Current	-	15	-	mA	-2dBm Output Power
I _{RX,LDO}	RX Mode Operating Current	-	7.8	-	mA	LDO Mode, maximum LNA gain
I _{RX,DCDC}	RX Mode Operating Current	-	4.1	-	mA	DCDC mode, maximum LNA gain
V _{OH}	Output High Voltage	VDD-0.3	-	VDD	V	-
V _{OL}	Output Low Voltage	VSS	-	VSS+0.3	V	-
V _{IH}	Input high level voltage	0.8*VDD	-	-	V	-
V _{IL}	Input low level voltage	-	-	0.2*VDD	V	-
SPI_rate	SPI rate	-	-	10	Mbps	-

5.3 RF Characteristics

Test conditions:

Frequency: 490MHz

纠错码 = 4/8

误包率 fl ~

Payload 长度 = 10Bytes

Table 5-4 General RF Characteristics (Frequency 490MHz)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
通用频率						
F _{op}	工作频率	-	137	-	187	MHz
		-	205	-	281	MHz
		-	408	-	550	MHz
		-	816	-	1110	MHz
F _{xtal}	晶振频率	-	-	32	-	MHz
R _S	晶体串联电阻	-	-	30	50	
C _{FOOT}	晶体外部电容	-	8	15	22	pF
C _{LOAD}	晶体负载电容	-	6	10	12	pF
F _{TOL}	初始频率容限	-	-	±10	-	ppm
BR	比特速率 (PAN3029 系列)	-	0.08	-	59.9	kbps
	比特速率 (PAN3060 系列)	-	0.5	-	59.9	kbps
发射器						
P _{LPWAN}	输出功率	-	-30	-	20	dBm
接收器						
RF_62.5 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-129 -137 -143	- - -	dBm
RF_125 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-126 -134 -140	- - -	dBm
RF_250 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-124 -132 -137	- - -	dBm
RF_500 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -134	- - -	dBm
RF_62.5 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -133 -139	- - -	dBm
RF_125 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-122 -130 -137	- - -	dBm
RF_250	RF 灵敏度, 长距离模式, 最高 LNA 增益,	SF = 7	-	-120	-	dBm

(DCDC)	使用分离的 RX/TX 通道 250 kHz 带宽	SF = 10	-	-128	-	
		SF = 12	-	-134	-	
RF_500 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7	-	-117	-	dBm
		SF = 10	-	-125	-	
		SF = 12	-	-130	-	

测试条件:

- 3 频率: 915MHz
- 3 纠错码 = 4/8
- 3 误包率 fl ~
- 3 Payload 长度 = 10Bytes

Table 5-5 General RF Characteristics (Frequency 915MHz)

符号	描述	条件	最小	典型	最大	单位
通用频率						
F _{op}	工作频率	-	137	-	187	MHz
		-	205	-	281	MHz
		-	408	-	550	MHz
		-	816	-	1110	MHz
F _{xtal}	晶振频率	-	-	32	-	MHz
R _S	晶体串联电阻	-	-	30	50	
C _{FOOT}	晶体外部电容	-	8	15	22	pF
C _{LOAD}	晶体负载电容	-	6	10	12	pF
F _{TOL}	初始频率容限	-	-	±10	-	ppm
BR	比特速率 (PAN3029 系列)	-	0.08	-	59.9	kbps
	比特速率 (PAN3060 系列)	-	0.5	-	59.9	kbps
发射器						
P _{LPWAN}	输出功率	-	-30	-	20	dBm
接收器						
RF_62.5 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -141	- - -	dBm
RF_125 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -132 -138	- - -	dBm
RF_250 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -135	- - -	dBm
RF_500 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-119 -127 -132	- - -	dBm
RF_62.5 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -140	- - -	dBm

RF_125 (DCDC)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7	-	-125	-	dBm
		SF = 10	-	-132	-	
		SF = 12	-	-137	-	
RF_250 (DCDC)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7	-	-120	-	dBm
		SF = 10	-	-129	-	
		SF = 12	-	-135	-	
RF_500 (DCDC)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7	-	-118	-	dBm
		SF = 10	-	-126	-	
		SF = 12	-	-130	-	

6 Frequency synthesizer

The chip integrates high-performance RF transceiver circuits and supports multiple Sub-1GHz frequency bands such as 150M/230M/400M/800M.

The chip contains a high-performance, low phase noise, fully integrated fractional frequency synthesizer. The frequency synthesizer is used to generate the LO frequency used by the down-conversion mixer in the receive mode. In the transmit mode, it is also used to directly generate the modulated RF carrier. The fractional frequency synthesizer provides excellent phase noise performance, frequency resolution less than 100Hz, and has low power consumption. The frequency synthesizer has a fast frequency stabilization function, which can complete the receive and transmit frequency stabilization in a very short time to optimize the chip power consumption.

7 Modem

The chip utilizes a modem that supports ChirpIoT™ transmission. This achieves an ultra-high communication link budget through superior spread-spectrum modulation and forward error correction technologies. Compared to traditional FSK-based modulation schemes, this increases the range and stability of the radio communication link.

The ChirpIoT™ modem significantly enhances interference immunity. Compared to FSK modulation, the ChirpIoT™ modem suppresses co-channel interference by up to 19dB, meaning that the interference signal energy can be 19dB higher than the ChirpIoT™ signal. This significantly ensures that the ChirpIoT™ signal can coexist easily with other modulation systems and be widely used in hybrid network communications.

7.1 Modulation parameters

The chip optimizes the transmission performance of the ChirpIoT™ signal by configuring different modulation parameters, achieving a balance between transmission rate, transmission distance, and transmission reliability. Specific parameters are as follows:

- Signal Bandwidth (BW)
- Spreading Factor (SF)
- Coding Rate (CR)
- Low Data Rate (LDR)

7.1.1 Signal Bandwidth

The PAN3029 supports four signal bandwidths: 500kHz, 250kHz, 125kHz, and 62.5kHz.

The PAN3060 supports three signal bandwidths: 500kHz, 250kHz, and 125kHz.

Different signal bandwidths can be selected through register configuration.

7.1.2 Spreading Factor

The PAN3029 supports eight different spreading factors, from SF5 to SF12, while the PAN3060 supports five different spreading factors, from SF5 to SF9. A larger spreading factor reduces the data payload rate.

7.1.3 Coding Rate

The chip uses forward error correction technology and supports four different coding rates: CR45, CR46, CR47, and CR48. This forward error correction technology corrects bit errors within a certain range, improving transmission reliability.

7.1.4 Low Rate Mode

Low rate mode is a communication mode of ChirpIoT™ modulation technology. It improves signal transmission reliability by reducing the number of valid bits of information carried per unit time.

7.2 Frame Structure

The frame structure of the ChirpIoT transmission signal consists of four parts: signal Preamble, Header, Payload, and CRC, as shown in the figure below.



Figure 7-1 PHY Frame Structure

8 MAC Design

8.1 Transmitter and receiver modes

The chip's transmitter has two unique operating modes, and the receiver has three unique operating modes.

Transmitter Operating Modes:

- Single-Packet Transmit Mode

- Continuous Transmit Mode

Receiver Operating Modes:

- Single-Packet Receive Mode

- Single-Packet Receive Mode with Timeout

- Continuous Receive Mode

8.2 Transmitter Mode

The MAC transmit mode has two modes: single-packet transmit and continuous transmit.

These modes are selected using the mode select register. After selecting TX mode in the mode select register, the state machine enters the TX ready state, waits for the FIFO to be filled with data, and then enters the TX transmit state.

When the MAC enters TX transmit mode, the power control module successively turns on analog circuits such as the LDO module, PLL module, and PA. It then sends a TX start signal to the modem module, initiating data transmission.

After data transmission is complete, the LDO module, PLL module, PA, and other analog circuits are sequentially turned off. Then send an IRQ signal to the MCU. After the MCU clears the IRQ, you can configure the REG_OPT_MODE register to enter the STB3 mode, ending a TX process.

8.2.1 Single packet transmit mode

Figure 8-1 shows the state flow diagram for single-packet transmission mode.

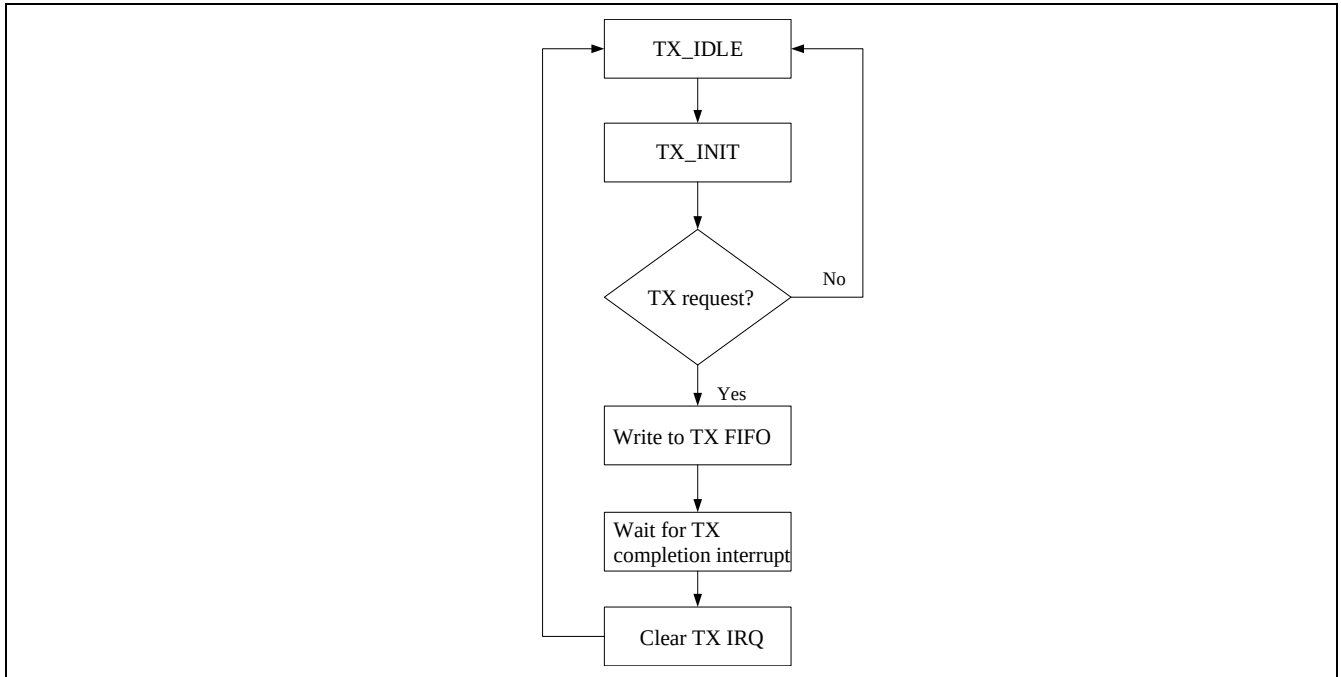


Figure 8-1 Single-Packet Transmit Mode State Flowchart

8.2.2 Continuous transmit mode

Figure 8-2 shows the state flow diagram for continuous transmit mode.

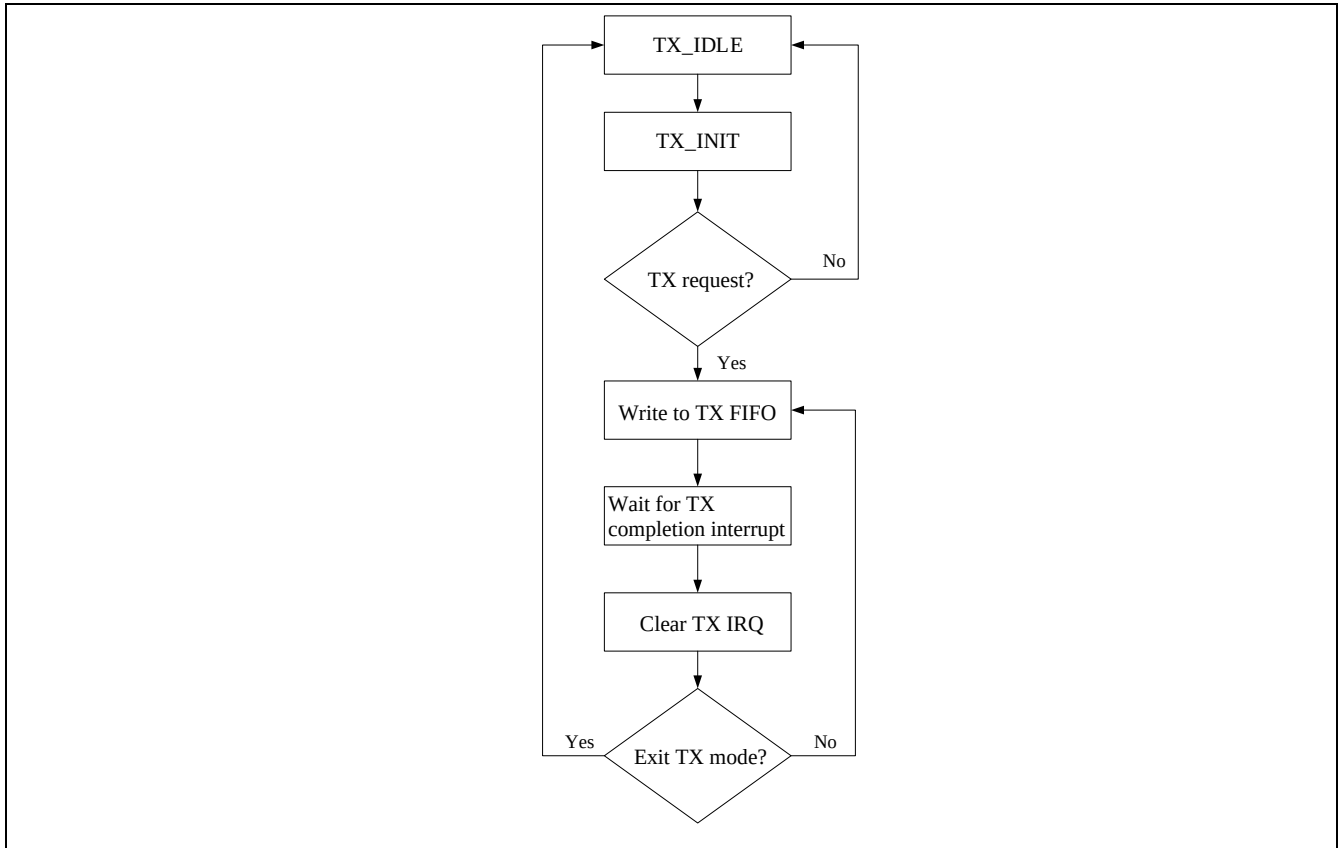


Figure 8-2 Continuous Transmit Mode Flowchart

8.3 Receiver Mode

The MAC receive mode includes single-packet receive, single-packet receive with timeout, and continuous receive modes, selected using a 2-bit mode select register.

When the MAC enters receive mode, the power control module successively turns on analog circuits such as the LDO and PLL, and simultaneously sends an RX start signal to the modem to begin receiving data.

After data reception is complete, the LDO, PLL, and other analog circuits are sequentially turned off. An IRQ signal is then sent to the MCU. After the MCU clears the IRQ, the REG_OPT_MODE register can be configured to enter STB3 mode, completing the RX process.

8.3.1 Single Packet Receive Mode

The state flow chart of the single packet receiving mode is shown in Figure 8-3.

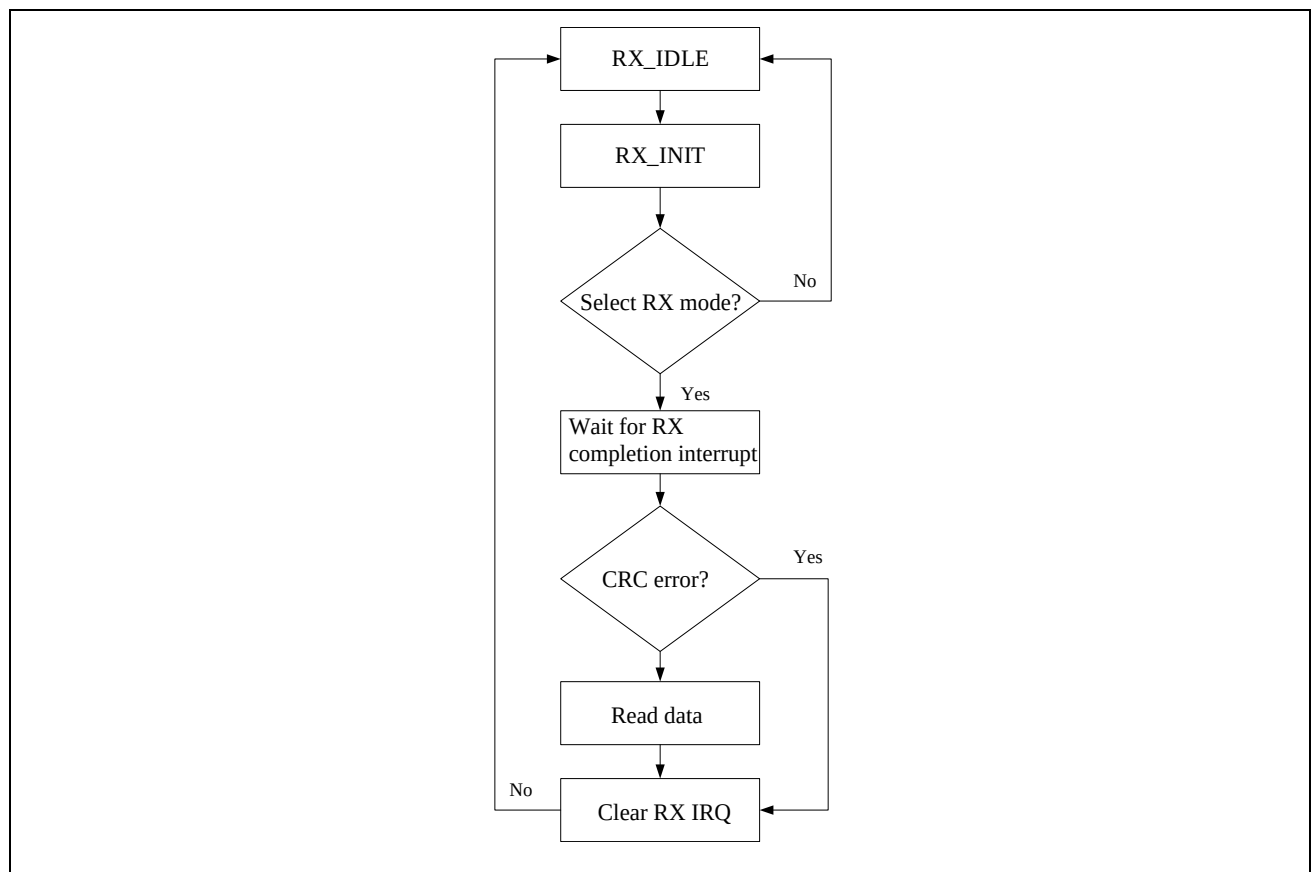


Figure 8-3 Single-Packet Receive Mode Flowchart

8.3.2 Single Packet Receive Mode with Timeout

Figure 8-4 shows the state flow diagram for single-packet receive mode with timeout.

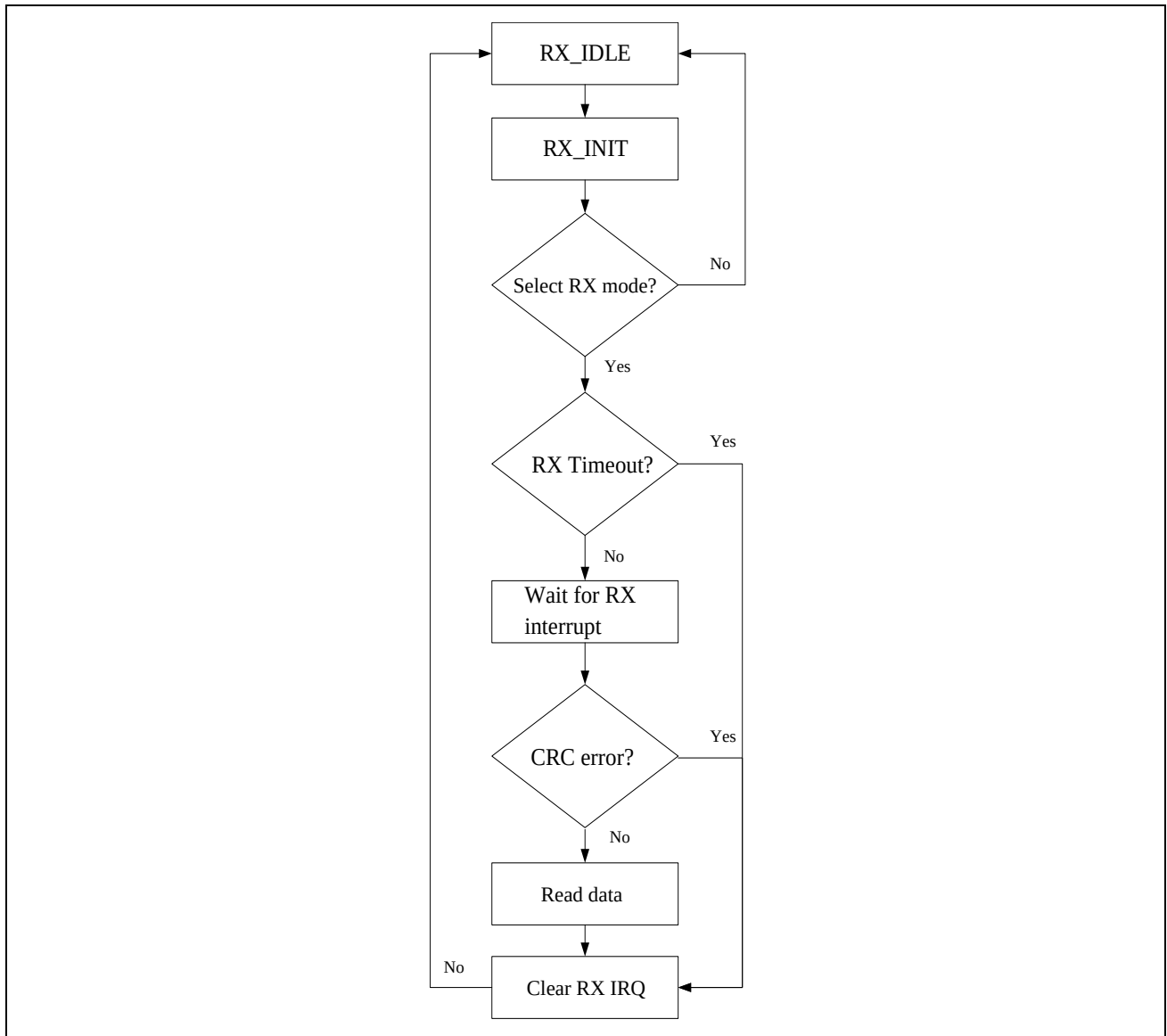


Figure 8-4 Single-Packet Receive Mode with Timeout Flowchart

8.3.3 Continuous Receive Mode

连续接收模式的状态流程图如图 8-5 所示。

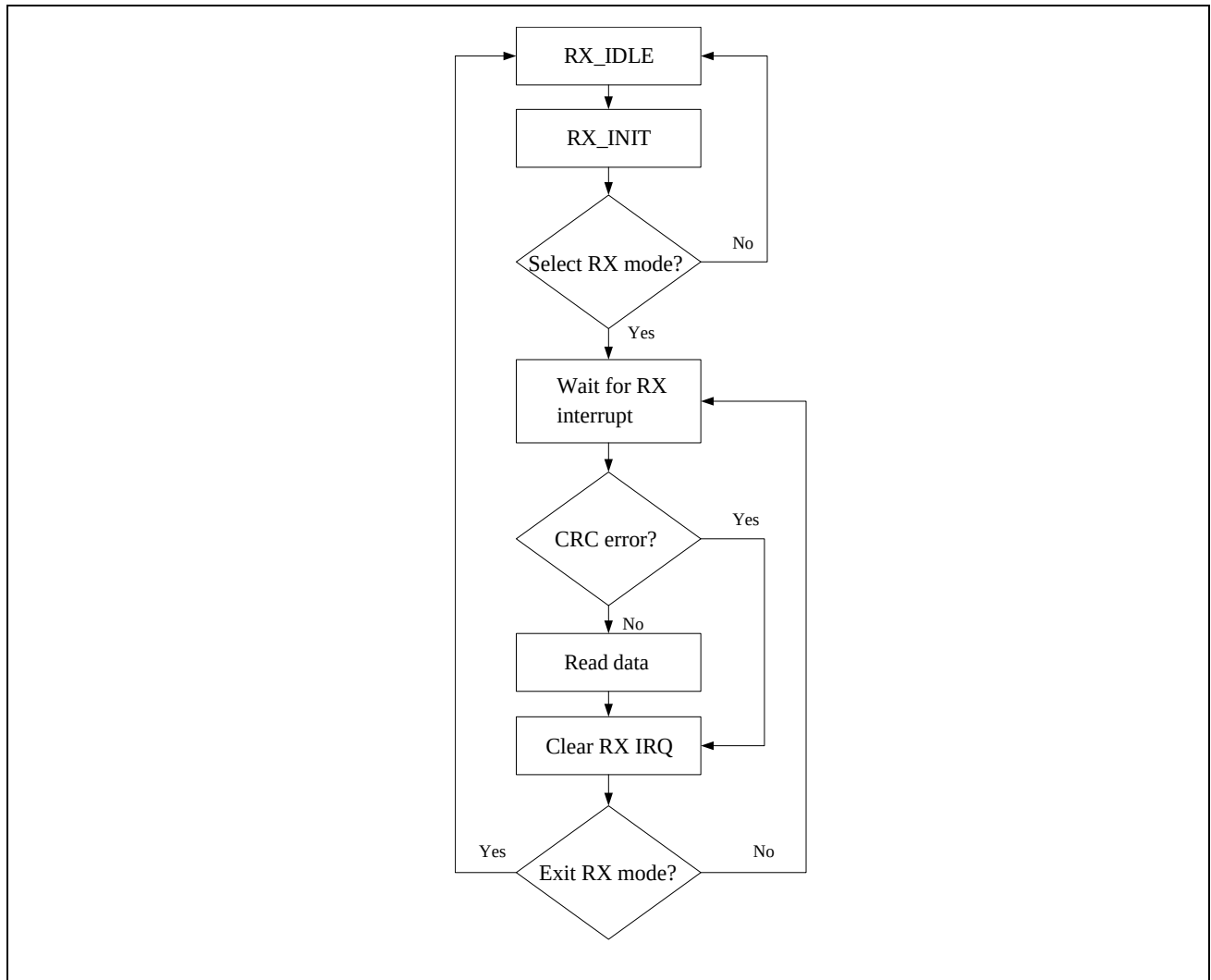


Figure 8-5 Continuous Receive Mode Flowchart

Continuous receive mode differs from single-packet receive mode and single-packet receive mode with timeout in that after data reception is complete, an IRQ signal is sent to the MCU. After the MCU clears the IRQ, the next data reception cycle begins. To exit continuous receive mode, the mode select register selects crystal oscillator output mode. Analog modules such as the LDO and PLL are then disabled, completing the continuous receive mode cycle.

8.4 IRQ Interrupt

All interrupts on this chip share a single external interrupt IO pin, IRQ, which is active high. The chip has seven interrupt types, as listed below. Interrupts can be enabled using the interrupt enable register, and their status can be read using the interrupt status register.

Interrupt Enable Register	Description
REG_MAPM_IRQ_MASK	IRQ Mask Enable in MAPM Mode 0: Generates MAPM IRQ interrupts 1: Does not generate MAPM IRQ interrupts
REG_RX_HMP_ERR_MASK	ECC Error IRQ Interrupt Mask 0: Does not generate IRQ interrupts for ECC errors 1: Generates IRQ interrupts for ECC errors
REG_RX_PLHD_DONE_MASK	Early Interrupt Mode Mask 0: Does not generate IRQ interrupts for early interrupt mode 1: Generates IRQ interrupts for early interrupt mode.
REG_RX_DONE_MASK	RX Completion IRQ Mask 0: Does not generate IRQ interrupts for RX completion 1: Generates IRQ interrupts for RX completion
REG_RX_PL_CRC_ERR_MASK	RX Payload CRC Error IRQ Mask 0: Does not generate IRQ interrupts for RX CRC errors 1: Generates IRQ interrupts for RX CRC errors.
REG_RX_TIMEOUT_MASK	RX Timeout IRQ Mask 0: Does not generate IRQ interrupts for RX timeouts 1: Generates an IRQ interrupt for Rx timeout
REG_TX_DONE_MASK	Tx completion IRQ mask 0: Does not generate an IRQ interrupt for Tx completion 1: Generates an IRQ interrupt for Tx completion

Interrupt Status Register	Description
REG_MDM_MAPM_IRQ	MAPM IRQ interrupt
REG_RX_HM_P_ERR_IRQ	ECC error IRQ interrupt
REG_RX_PLHD_DONE_IRQ	Early interrupt mode IRQ interrupt
REG_RX_DONE_IRQ	RX completion IRQ interrupt
REG_RX_PL_CRC_ERR_IRQ	RX payload CRC error IRQ interrupt
REG_RX_TIMEOUT_IRQ	RX timeout IRQ interrupt
REG_TX_DONE_IRQ	TX completion IRQ interrupt

8.5 CAD-IRQ Interrupt

The chip's CAD-IRQ interrupt pin reuses GPIO11. To use it, you need to configure GPIO11 hardware to CAD-IRQ output mode.

Register Name	Register Location	Description
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD Function Selection When REG_OUT_FUNC_SEL_CAD is 1, the GPIO functions as a normal IO pin, and its output can be configured using the PAD_OUT_REG register. When REG_OUT_FUNC_SEL_CAD is 0, the GPIO output functions as a CAD pin.
PAD_OUT_REG_CAD	page0, 0x68, bit3	Controls the GPIO11 output register.

9 Interface Design

The chip supports reading and writing registers and transmit/receive FIFOs via four-wire SPI, three-wire SPI, and I2C. By default, four-wire SPI and I2C are both supported. Three-wire SPI requires configuration via the SPI3_EN_REG register.

Note: The address is a special area. The register address in this area is the entry address for this special area, and its operation differs from that of regular registers. Please refer to the section for details on the operation.

9.1 Four-wire SPI

The chip implements an SPI bus slave, used to read and write registers and FIFOs. The SPI bus is a four-wire system consisting of:

- SCK (clock)
- CSN (chip select, active low)
- MOSI (data input)
- MISO (data output)

SCK, CSN, and MOSI are controlled by the master, while MISO is controlled by the slave.

Communication begins when CSN is pulled low and ends when CSN is pulled high. The master sends data via MOSI and receives data via MISO. Data is generated on the falling edge of SCK and sampled on the rising edge.

The information transmitted by the master consists of two parts: the address byte and the data byte. The first 7 bits of the address byte are the address bit (addr); the last bit is the read/write bit (wr). This bit is set to 1 during a write operation and to 0 during a read operation.

SPI has three transmission modes:

Single: Single-byte transmission mode. The message is only 2 bytes long. The master sends the address byte via MOSI. If it is a write operation, the master continues to send data bytes via MOSI. If it is a read operation, the master reads the data bytes replied by the slave on MISO.

Burst: Burst continuous transmission mode. When the message is longer than 2 bytes, the address byte is followed by several data bytes. There is no need to add address bytes between data bytes. The slave automatically increments the address between each data byte. The CSN signal is pulled high after the last data byte and remains low during the rest of the transmission.

FIFO: FIFO read/write mode. This mode supports both single-byte and continuous transmission. The transmission rules are the same as those in Single and Burst modes. The difference is that the address bit addr in the address byte can only be configured as 0x01, and the slave does not increment the address between data bytes.

In addition, the REG_OUT_PAD_MODE register can be configured to set MISO to input high-impedance state to implement a 4-wire SPI one-master-multiple-slave connection architecture.

REG_OUT_PAD_MODE	0 (dec)	1 (dec)	2 (dec)	3 (dec)	3 (dec)
MISO_OUT	MISO_3V	0	1	CSN_IN=1 Output 0	CSN_IN=0 Output MISO_3V
MISO_DIEN	0	0	0	0	0
MISO_OE	1	1	1	CSN_IN=1 0	CSN_IN=0 1
MISO_PUEN	0	0	0	0	0
MISO_PDEN	0	0	0	0	0
NOTE	Normal output	Output 0	Output 1	CSN_IN=1 No output	CSN_IN=0 Normal output

9.1.1 Four-wire SPI write timing

The SPI write timing is as follows:

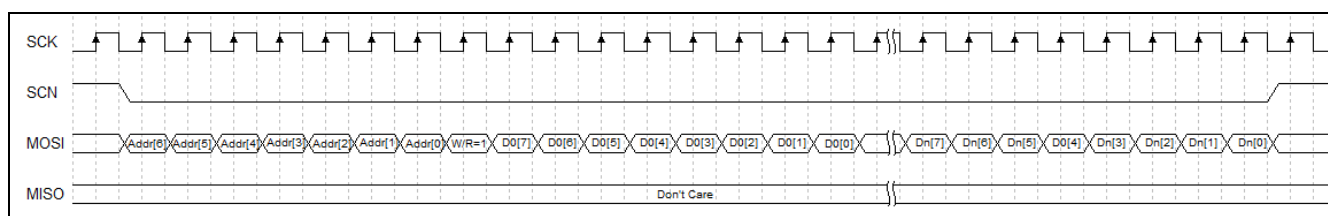


Figure 9-1 SPI Write Timing

9.1.2 Four-wire SPI read timing

The SPI read timing is as follows:

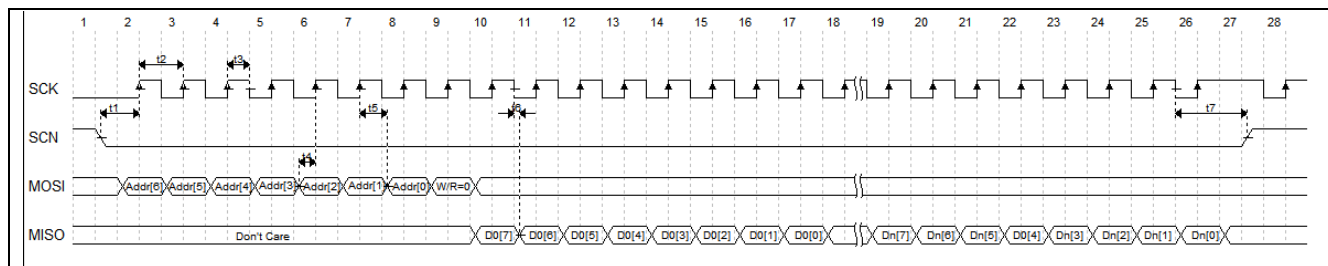


Figure 9-2 SPI Read Timing

9.1.3 Four-wire SPI timing requirements

All SPI timing requirements in the table below are as follows:

Symbol	Description	Min	Typ	Max	Unit
t1	SCN falling edge to SCK rising edge	32	-	-	ns
t2	SCK clock period	100	-	-	ns
t3	SCK high time	50	-	-	ns
t4	MOSI to SCK clock setup time	7	-	-	ns
t5	MOSI to SCK clock hold time	7	-	-	ns
t6	SCK falling edge to MISO delay	0	-	15	ns
t7	SCK to SCN rising edge time	0	-	150	ns

9.2 Three-wire SPI

By default, the chip supports four-wire SPI and I2C. If you require three-wire SPI, you must configure the SPI3_EN_REG register after power-up to enable three-wire SPI. Three-wire SPI signals include:

- SCK (clock)
- CSN (chip select signal, active low)
- MOSI (data input/output)

9.2.1 Three-wire SPI write timing

SPI 写时序如下:

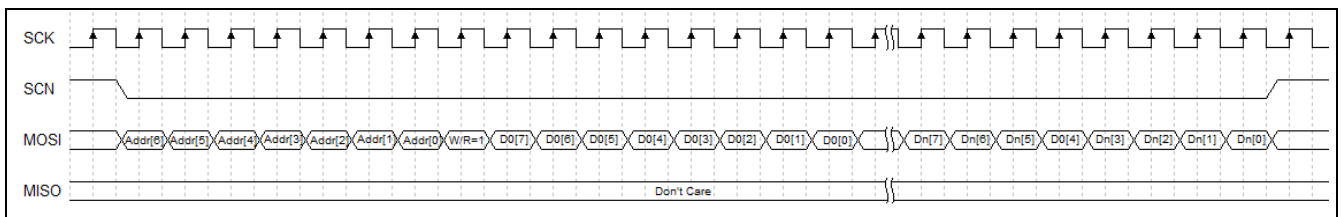


Figure 9-3 3-Wire SPI Write Timing

9.2.2 Three-wire SPI read timing

The SPI write timing is as follows:

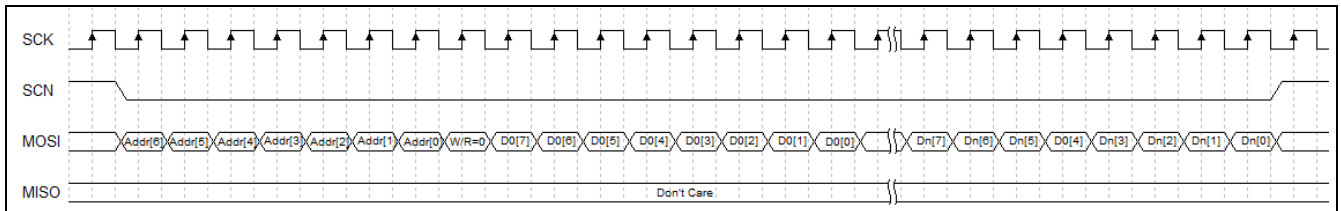


Figure 9-4 3-Wire SPI Read Timing

9.2.3 Three-wire SPI timing requirements

Please refer to Section 9.1.3.

9.3 I2C

The chip supports I2C by default. The I2C signals SCL and SDA multiplex the SPI signals SCK and MOSI.

Note: When using I2C, the CSN signal must be high.

The I2C device address is 7 bits wide and contains 0x72.

The timing diagram of the I2C start signal is as follows:

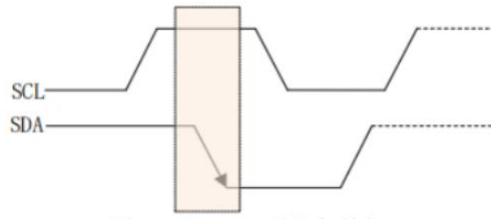


Figure 9-5 I2C Start Signal

The I2C stop signal timing diagram is as follows:

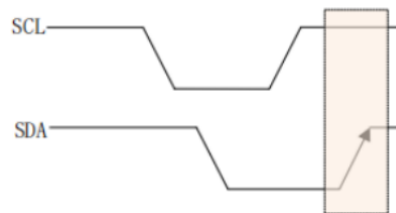


Figure 9-6 I2C Stop Signal

9.3.1 I2C Write Timing

The I2C write timing is as follows:

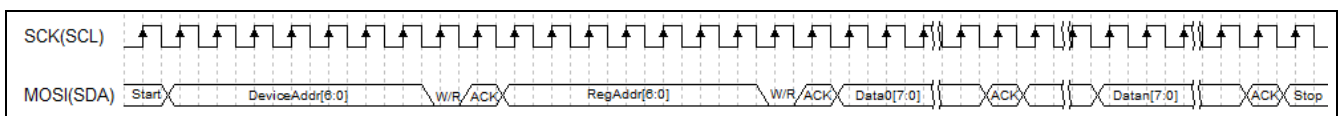


Figure 9-7 I2C Write Timing

9.3.2 I2C Read Timing

The I2C read timing is as follows:

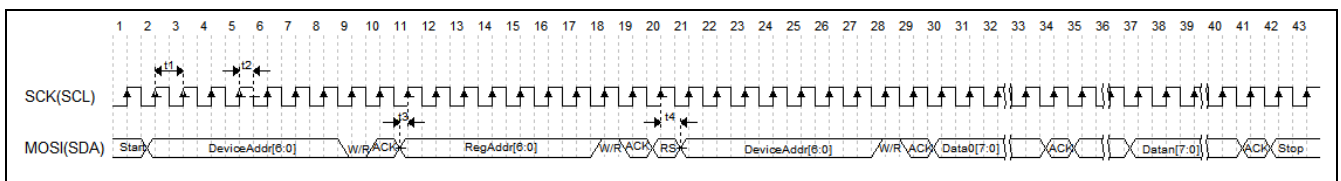


Figure 9-8 I2C Read Timing

9.3.3 I2C Timing Requirements

Symbol	Description	Min	Typ	Max	Unit
t1	SCL clock period	200	-	-	ns
t2	SCL high time	100	-	-	ns
t3	SDA to SCL clock setup time	12	-	-	ns
t4	SDA to SCL clock hold time	12	-	-	ns

9.4 FIFO

The chip features a 256-byte FIFO for storing data transmitted by the TX module and decoded data from the RX module.

The FIFO consists of a single-port RAM and can only store and read a single packet of data. If a packet already exists in the FIFO, it must be read before writing to the previous packet; otherwise, the previous packet in the FIFO will be overwritten.

In STB3 and later operating modes, the FIFO can be read and written via a modem, SPI, or I2C.

The FIFO's read and write address is 0x01, which can be used to read and write the FIFO contents via SPI or I2C.

9.5 GPIO

The chip supports multiple GPIO input and output control, which can be controlled through register configuration. For register configuration, please refer to Section 12.4. Each GPIO port can be configured as an internal pull-up/pull-down input, a high-impedance input (floating input), or a push-pull output (CMOS output). The configuration methods are shown in Table 9-1.

Table 9-1 GPIO port mode configuration

DIEN	OE	OUT	PUEN	PDEN	IO 方向	IO 状态
0	1	0/1	0	0	推挽输出	0/1
0	0	x	0	0	开漏输出	高阻
1	0	x	0	0	输入	高阻
1	0	x	1	0	输入	上拉
1	0	x	0	1	输入	下拉

注意：芯片中 和 的输出模式可被复用为实现特殊功能。

9.5.1 External PA Control GPIO

GPIO10 有两种配置输出模式：普通 IO 和 PA 控制管脚。通过寄存器 REG_OUT_FUNC_SEL_PA 可配置 GPIO10 的不同模式。

当 GPIO10 配置为普通 IO 时，可以输出高低电平，与其他 GPIO 无异。

当 GPIO10 被配置为 PA 控制时，并且外控 PA 模式通过寄存器配置为开启状态，GPIO10 可作为外部 PA 的使能控制引脚，即 GPIO10 仅在发射模式下发送数据时输出高电平，其他时间输出低电平。

寄存器名称	寄存器位置	描述
REG_OUT_FUNC_SEL_PA	page0, 0x5e, bit7	GPIO10 功能选择 当 REG_OUT_FUNC_SEL_PA 为 1, GPIO10 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG_PA 配置; 当 REG_OUT_FUNC_SEL_PA 为 0, GPIO10 输出为外置 PA 功能管脚。
PAD_OUT_REG_PA	page0, 0x68, bit2	控制 GPIO10 输出的寄存器。 当 REG_OUT_FUNC_SEL_PA 为 1, GPIO10 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG_PA 配置。

9.5.2 CAD monitoring GPIO

GPIO11 有两种配置模式：普通 IO 和 CAD 检测管脚。通过寄存器 REG_OUT_FUNC_SEL_CAD 可配置 GPIO11 的不同模式。

当 GPIO11 配置为普通 IO 时，可以输出高低电平，与其他 GPIO 无异。

当 GPIO11 被配置为 CAD 信号检测中断输出时，并且 CAD 功能也通过寄存器配置为开启状态，以及芯片处于 RX 状态时，当接收到与当前配置的带宽和扩频因子参数相同的信号时，GPIO11 就会产生一个连续稳定的高电平，指示芯片搜寻到了空中信号，该高电平直到信号消失才会拉低；其他情况和模式，GPIO11 都输出低电平。具体 CAD 配置使用可参考 章节。

芯片 CAD-IRQ 中断引脚复用了 GPIO11 引脚，使用时需要将 GPIO11 硬件配置成 CAD-IRQ 输出模式。

寄存器名称	寄存器位置	描述
REG_OUT_FUNC_SEL_CAD	page0, 0x5e, bit6	CAD 功能选择 当 REG_OUT_FUNC_SEL_CAD 为 1, GPIO 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG 配置; 当 REG_OUT_FUNC_SEL_CAD 为 0, GPIO 输出为 CAD 功能管脚。
PAD_OUT_REG_CAD	page0, 0x68, bit3	控制 GPIO11 输出的寄存器。 当 REG_OUT_FUNC_SEL_CAD 为 1, GPIO11 为普通的 IO, 其输出可由寄存器 PAD_OUT_REG_CAD 配置。

10 ChirpIoT special function description

10.1 RSSI Function

10.1.1 RSSI indication of ChirpIoT signal

芯片支持信号能量统计功能。芯片接收到 ChirpIoT 数据时，芯片内部逻辑会统计当前数据包的信号强度 RSSI 值，但数据包接收完成后，MCU 可通过 SPI 接口读取寄存器 RSSI_R32_HOLD 或者 RSSI_RBW_HOLD 来获取当前数据包的 RSSI 值。

10.1.2 Channel detection RSSI indication

RSSI 功能同时支持检测信道时的能量统计指示。芯片在进入 RX 状态后，通过配置寄存器 REG_RSSI_PRE_RD，可进行一次 RSSI 能量统计，而后通过寄存器 RSSI_R32_HOLD 可读取当前信号 RSSI 能量。

REG_RSSI_PRE_RD 先配置为 0，再配置为 1，可以更新一次 RSSI 值，寄存器位置如下：

寄存器名称	寄存器位置	描述
REG_RSSI_PRE_RD	page2, 0x06, bit2	检测信道 RSSI 读取开关，由 0 配置为 1 时更新一次 RSSI 值。

10.2 Signal-Noise Energy Statistics Function

芯片支持信号噪声统计功能。芯片接收到 ChirpIoT 数据时，芯片内部逻辑会计算当前数据包的信号能量和噪声能量，但数据包接收完成后，MCU 可通过 SPI 接口读取寄存器 SIG_POW_AVG 的信号能量值和寄存器 NOI_POW_AVG 的噪声能量值，然后 MCU 可通过标准公式计算得到该数据包的信噪比情况。

10.3 Channel Activity Detection (CAD)

由于 ChirpIoT™信号具有优异的接收性能，在低于接收机底噪声的情况下仍然能保证正常的信号接收解调，在这样的情况下，仅使用信号强度指示 RSSI 来判断信道是否被占用的方式已经无法满足需求，因此，芯片专门设计了信道活动检测器来检测 ChirpIoT™信号，保证能在低于接收机底噪声的情况下也能发现信道被其他设备发送的 ChirpIoT™信号占用。

信道活动检测器的检测功能分成两个部分，一个是 ChirpIoT™信号前导段检测，一个是 ChirpIoT™信号数据段检测。

10.3.1 ChirpIoT signal preamble detection

如图 10-1 所示:

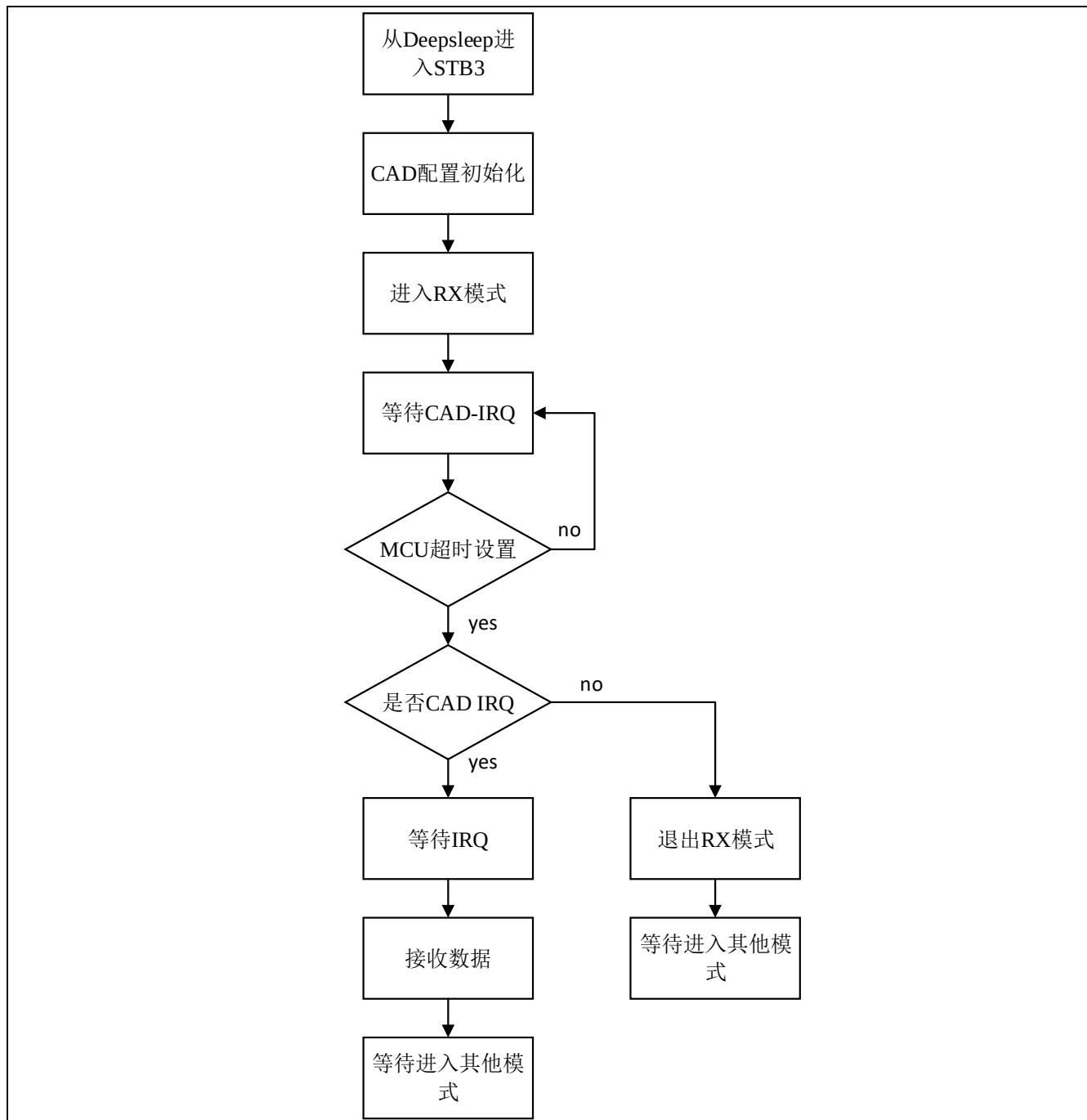


Figure 10-1 Channel Activity Detection (CAD)

芯片支持 CAD-IRQ 中断, 在进入 RX 模式后, 芯片会检测信道中是否存在 ChirpIoT™ 的前导信号, 如果存在则将 CAD-IRQ 置高, 外部 MCU 可以通过在一定时间内检测 CAD-IRQ 信号是否拉高来判断信道中是否存在 ChirpIoT™ 的前导信号。如存在则可以继续等待数据接收的 IRQ 中断, 而后进行数据接收; 如不存在 ChirpIoT™ 的前导信号,

则快速退出 RX 模式，等待进入其他模式命令。

另外，为了减少 CAD-IRQ 的误触发，增加了 CAD-IRQ 的滤波功能，通过配置寄存器可启动该功能。

地址	寄存器	BIT	推荐值	读写	说明
[1][0x25]	REG_DEC_VAL_THRE	1:0	0x3	W/R	CAD 检测阈值，规定连续接收到几个 preamble 之后 CAD 拉高。

10.3.2 ChirpIoT signal data segment detection

为了避免错过 ChirpIoT 信号的前导信号而导致实际接收过程中无法实现有效 CAD 检测，芯片还同时支持非 ChirpIoT 前导信号的检测，即 ChirpIoT 数据段信号检测功能。通过配置寄存器 REG_SIGNAL_POWER_THRED、REG_DEC_PLD_THRE、REG_DEC_PLD_EN 可实现启动数据段信号检测。

寄存器配置说明：

地址	寄存器	BIT	推荐值	读写	说明
[1][0x35]	REG_SIGNAL_POWER_THRED	7:4	4'b1110	W/R	CAD 有效信号的阈值，只有能量大于此值才能进入 CAD 有效信号的判断。
	REG_DEC_PLD_THRE	3:2	2'b10	W/R	CAD 有效信号个数阈值，只有接收到满足条件的此数值个数的 preamble，才能拉高 CAD。
	REG_DEC_PLD_EN	1	1'b0	W/R	CAD payload 检测使能 0: disable 1: enable

注意：前导段和数据段的 CAD 检测可以同时打开，确保不同阶段数据可检测判断。

10.4 MAPM Function

在 ChirpIoT 星状组网使用场景中，为了保证节点端设备能收到网关发送的数据，网关端会使用超长 Preamble 进行发送，如 preamble 占用时间 1 秒。在这种情况下，所有节点端设备在收到 preamble 后都会一直接收直到 payload 阶段显示节点网络地址，此时，节点端设备的软件可以比较收到的网络地址和节点设备自己地址，如不同，则停止接收，降低功耗，如满足地址接收条件则继续接收。这种方案存在几个问题：

- ① 不满足地址接收条件的节点端设备，直到 payload 阶段才能停止接收，浪费大量功耗
- ② 满足地址接收条件的节点端设备也需要收满超长的 preamble 才能进入 payload 接收阶段，也较浪费功耗

为解决以上两个功耗问题，芯片增加新的帧结构模式 Multiple Address Preamble Mode (MAPM)。该帧结构模式是在 Preamble 阶段插入特殊地址，这样节点端在 Preamble 阶段就可以根据地址来提前判断是否满足地址接收条件，已解决①问题。另外，为了

解决第②问题，在特殊地址中增加可变计数器，表明后续还有多少 Preamble 长度，满足地址接收条件的节点端设备可提前进入低功耗休眠状态，后根据该计数器判断需要提前多久在进入接收状态，保证数据正常接收，以便进一步节省满足地址接收条件的节点端设备功耗。

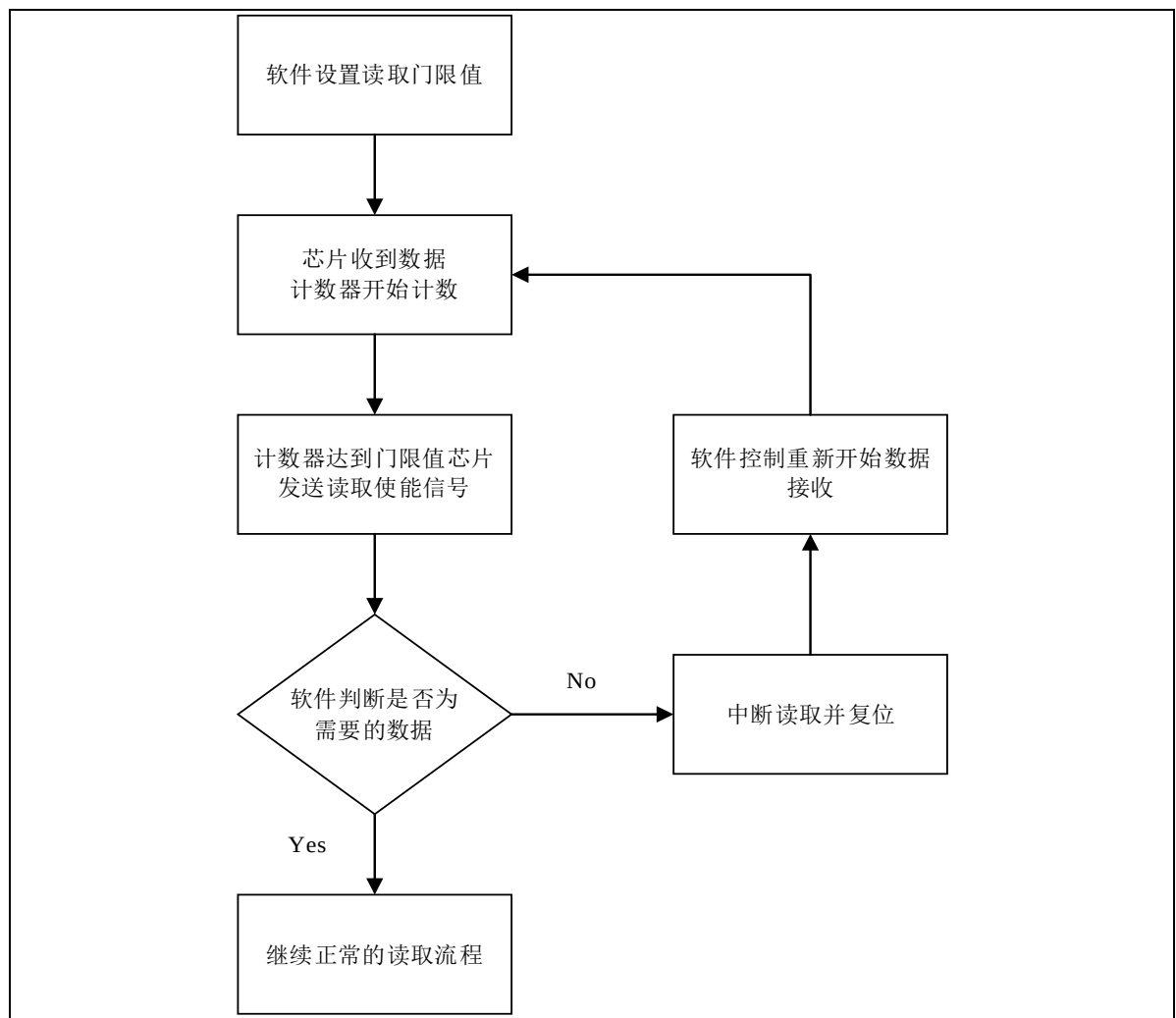
10.5 eFuse Function

芯片支持 82Bytes 的 eFuse 空间区域供客户使用，可用于存储掉电不丢失的数据。

10.6 Pre-interrupt Function

芯片提前中断功能是在芯片读取一帧数据的过程中，查看已经解出来的数据，判断是不是自己想要的，再决定继续读取还是放弃这帧数据。

流程图如下所示：



10.7 Intelligent SF Identification

芯片为实现轻量化网关设备，设计了智能化 SF 识别硬件电路，可完成在同一个频点根据实际信道中的 ChirpIoT™的 SF 模式，智能化识别，并自动配置芯片 SF 参数，达到接收不同 SF 信号数据的目的。

具体实现流程如图 10-2 所示：

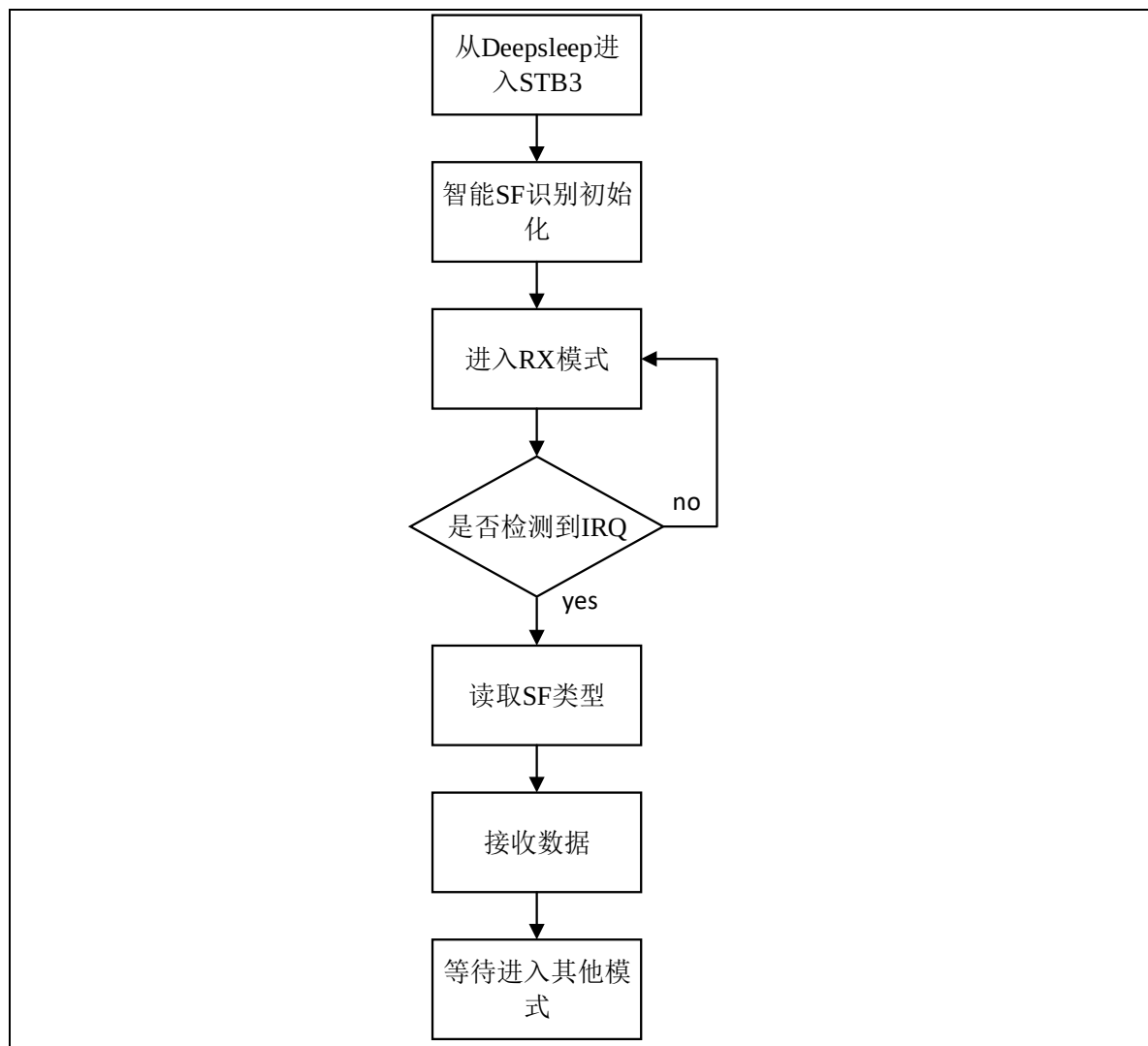


Figure 10-2 Intelligent SF Identification

11 Operation Modes Description

11.1 Overview

芯片一共有 7 种模式：DeepSleep 模式、Sleep 模式、STB1 模式、STB2 模式、STB3 模式、TX 模式和 RX 模式。

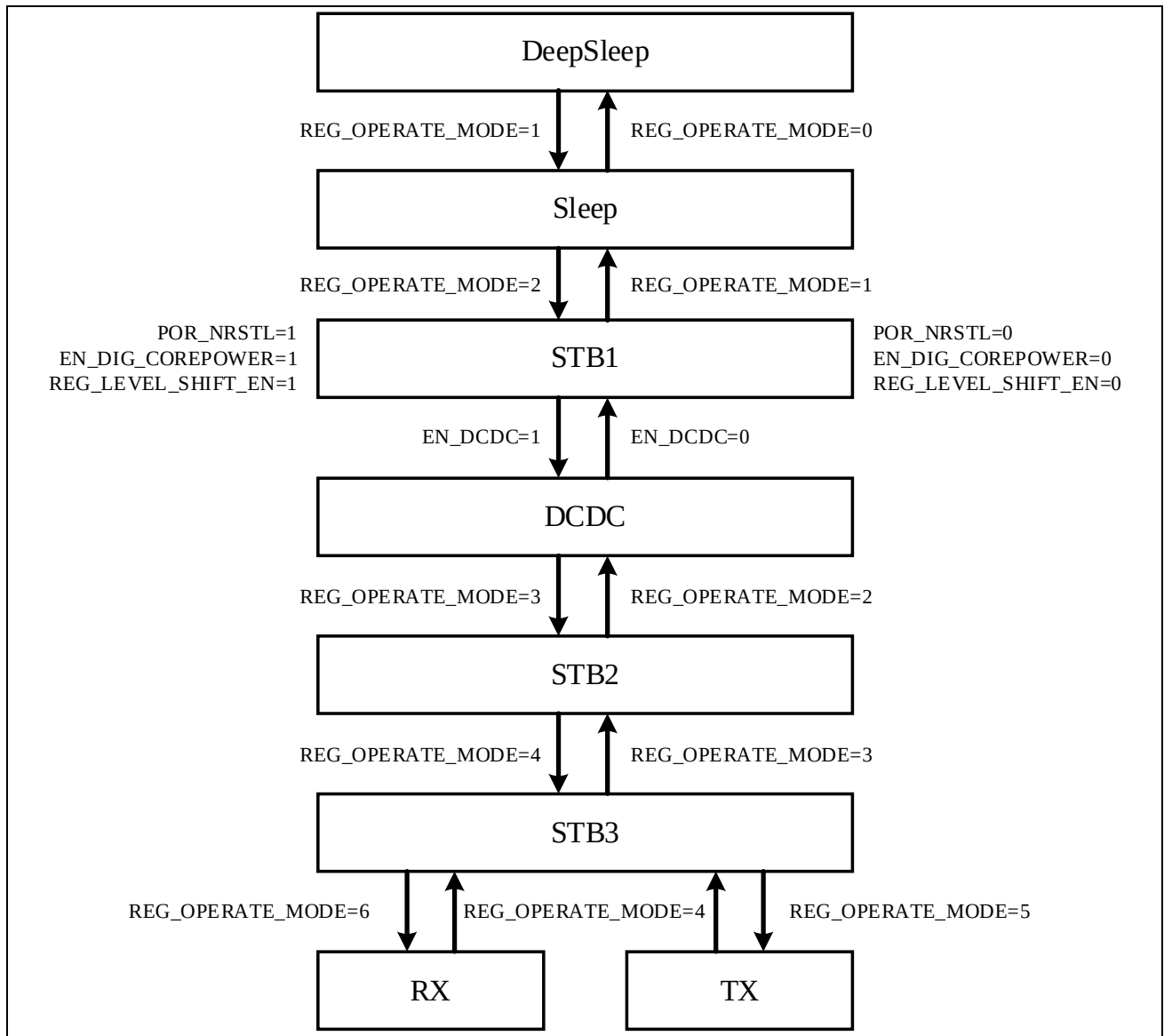


Figure 11-1 Operating Mode Setting Process

芯片上电后有七种状态，包括深度睡眠模式(DeepSleep)、睡眠模式(Sleep)、LDO 工作模式(STB1)、OSC 工作模式(STB2)、OSC 输出模式(STB3)、TX 模式以及 RX 模式。所有状态可通过配置寄存器进行切换。

注：如果芯片外围电路不支持 DCDC，可不配置 EN_DCDC，直接从 STB1 进入 STB2。芯片电路可参考第 13 章。

11.2 Working Modes Description

11.2.1 Deep Sleep Mode

该模式下 3V 逻辑区域工作，除 3V 逻辑区域寄存器可配置，其余模块不可操作，保持芯片功耗最小。

11.2.2 Sleep Mode

该模式下使能低功耗 LDO，芯片功耗较低，且可支持所有寄存器保电，但禁止通过 SPI/I2C 配置除 3V 逻辑区域以外的其他寄存器。

11.2.3 STB1 Mode

该模式高性能 LDO 开始工作，低压区域逻辑使能。

11.2.4 DCDC Mode

在 STB1 状态下可通过给 EN_DCDC 寄存器写入高电平来打开芯片的 DCDC 功能。

注意：DCDC 模式需要相关外围电路支持，请参考第 13 章。

11.2.5 STB2 Mode

该模式打开 OSC 晶振，OSC 开始工作维持震荡但不输出给其他模块，功耗相对较小。

11.2.6 STB3 Mode

该模式下 OSC 时钟输出给各模块，芯片开始正常工作。

11.2.7 TX Mode

该模式已打开所有 TX 发射相关的模块，可实现包数据的编码和发送。

11.2.8 RX Mode

该模式已打开所有 RX 接收相关的模块，等待包数据的接收和解码。

12 Registers

芯片的寄存器分为 4 页，每页共 128 个寄存器地址，通过寄存器 0x00 的 REG_PAGE_SEL 进行寄存器页选择，其中地址 0x00~0x05 的寄存器在任意一个寄存器页都可以实现操作。下表描述了芯片每个寄存器的位置和功能。

注意：

- 禁止对 *Reserved* 部分的寄存器进行读写操作。
- 所有低压区的寄存器仅支持在 STB1、STB2、STB3、Tx 和 Rx 模式操作。

12.1 System Configuration

page	地址	寄存器	BIT	推荐值	读写	说明
0/1/2/3	0x00	REG_SOFT_RST	7	0x00	W/R	软复位，除 1.2V 配置寄存器外的所有 1.2V 逻辑复位 0: 复位 1: 不复位
		Reserved	6:2		W/R	Reserved
		REG_PAGE_SEL	1:0		W/R	0: 选择 Page0 1: 选择 Page1 2: 选择 Page2 3: 选择 Page3
	0x01	FIFO READ/ WRITE ACCESS POINT	7:0	-	W/R	FIFO 读写地址

12.2 Mode Configuration

page	地址	寄存器	BIT	推荐值	读写	说明
0/1/2/3	0x02	Reserved	7:3	0x00	W/R	Reserved
		REG_OPERATE_MODE	2:0		W/R	0: 深度睡眠模式 1: 睡眠模式 2: STB1 模式(LDO 工作) 3: STB2 模式(OSC 工作) 4: STB3 模式(OSC 输出) 5: Tx 模式 6: Rx 模式
	0x04	Reserved	7:6	0x06	W/R	Reserved
		EN_LS_3V	5		W/R	1: 电平转换使能
		POR_RSTL	4		W/R	1.2V 所有配置寄存器复位 0: 复位 1: 不复位
		Reserved	3:0		W/R	Reserved

12.3 MAC Configuration (Low Voltage Area Register)

page	地址	寄存器	BIT	推荐值	读写	说明
0	0x54	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_FIFO_FULL	2		W/R	0: TX FIFO 未 1: TX FIFO 已
		REG_TX_FIFO_FULL_SEL	1		W/R	0: select fifo wr done 1: select reg tx fifo full
		Reserved	0		W/R	Reserved
	0x58	REG_RX_TIMEOUT_IMMED	7	0x0F	W/R	1: Rx 超时 IRQ 立即生成
		REG_MAPM_IRQ_MASK	6		W/R	MAPM 模式下的 IRQ MASK 使能 0: 不产生 MAPM 的 IRQ 中断 1: 产生 MAPM 的 IRQ 中断
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_MASK	4		W/R	提前中断模式掩码 0: 不产生提前中断模式的 IRQ 中断 1: 产生提前中断模式的 IRQ 中断
		REG_RX_DONE_MASK	3		W/R	rx 完成 irq 掩码 0: 不产生 rx 完成的 IRQ 中断 1: 产生 rx 完成的 IRQ 中断
		REG_RX_PL_CRC_ERR_MASK	2		W/R	Rx 有效负载 crc 错误 irq 掩码 0: 不产生 Rx 有效负载 crc 错误的 IRQ 中断 1: 产生 Rx 有效负载 crc 错误的 IRQ 中断
		REG_RX_TIMEOUT_MASK	1		W/R	Rx 超时 irq 掩码 0: 不产生 Rx 超时的 IRQ 中断 1: 产生 Rx 超时的 IRQ 中断
		REG_TX_DONE_MASK	0		W/R	Tx 完成 irq 掩码 0: 不产生 Tx 完成的 IRQ 中断 1: 产生 Tx 完成的 IRQ 中断
	0x5b	Reserved	7:3	0x03	W/R	Reserved
		REG_RX_DONE_RST_TEST	2		W/R	1: 一旦 RX 完成, modem 即刻复位 0: IRQ 清零且 RX 完成后 modem 才复位
		REG_RX_CR_ERR_RST	1		W/R	1: rx 码率错误复位使能
		REG_RX_HEADER_ER_RST	0		W/R	1: rx 包头错误复位使能
	0x5d	Reserved	7:1	0x00	W/R	Reserved
		REG_2POINT_CAL_EN	0		W/R	两点 cal 使能, 高电平有效
	0x6c	EFUSE_WR_IRQ	7	0x00	R	eFuse 读写完中断 1: 高电平说明读写完成
		REG_MDM_MAPM_IRQ	6		R	MAPM 中断状态寄存器 1: MAPM IRQ; 0: 非 MAPM IRQ
		Reserved	5		W/R	Reserved
		REG_RX_PLHD_DONE_IRQ	4		W/R	写 1 清零提前中断
		REG_RX_DONE_IRQ	3		W/R	写 1 清零 rx done irq
		REG_RX_PL_CRC_ERR_IRQ	2		W/R	写 1 清零 rx payload crc error irq

		REG_RX_TIMEOUT_IRQ	1		W/R	写 1 清零 rx timeout irq
		REG_TX_DONE_IRQ	0		W/R	写 1 清零 tx done irq
	0x6D	REG_MAPM_PN_CNT	7:0	0x00	R	MAPM 模式的 preamble 数量计数器
	0x6E	REG_MAPM_ADDRX	7:0	0x00	R	MAPM 模式的 addrx

12.4 GPIO interface configuration (low voltage area register)

page	地址	寄存器	BIT	推荐值	读写	说明
0	0x5e	REG_OUT_FUNC_SEL_PA	7	0x00	W/R	1: 作为普通 IO 使用 0: 作为外置 PA 使用
		REG_OUT_FUNC_SEL_CAD	6		W/R	1: 作为普通 IO 使用 0: 作为 CAD 引脚使用
		Reserved	5:0		W/R	Reserved
	0x5f	GPIO_PDEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 下拉使能 GPIO_PDEN LSB[0]: GPIO0 下拉使能 其他配置 Reserved
	0x60	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PDEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 下拉使能 GPIO_PDEN MSB[2]: GPIO10 下拉使能 其他配置 Reserved
	0x61	GPIO_PUEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 上拉使能 GPIO_PDEN LSB[0]: GPIO0 上拉使能 其他配置 Reserved
	0x62	Reserved	7:4	0x00	W/R	Reserved
		GPIO_PUEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 上拉使能 GPIO_PDEN MSB[2]: GPIO10 上拉使能 其他配置 Reserved
	0x63	GPIO_DIEN LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输入使能 GPIO_PDEN LSB[0]: GPIO0 输入使能 其他配置 Reserved
	0x64	Reserved	7:4	0x00	W/R	Reserved
		GPIO_DIEN MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输入使能 GPIO_PDEN MSB[2]: GPIO10 输入使能 其他配置 Reserved
	0x65	GPIO_OE LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输出使能 GPIO_PDEN LSB[0]: GPIO0 输出使能 其他配置 Reserved
	0x66	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OE MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输出使能 GPIO_PDEN MSB[2]: GPIO10 输出使能 其他配置 Reserved
	0x67	GPIO_OUT LSB	7:0	0x00	W/R	GPIO_PDEN LSB[3]: GPIO3 输出 GPIO_PDEN LSB[0]: GPIO0 输出 其他配置 Reserved
	0x68	Reserved	7:4	0x00	W/R	Reserved
		GPIO_OUT MSB	3:0		W/R	GPIO_PDEN MSB[3]: GPIO11 输出

						GPIO_PDEN MSB[2]: GPIO10 输出 其他配置 Reserved
	0x74	Reserved	7:6	-	R	Reserved
		GPIO_IN LSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO3 输入 GPIO_PDEN LSB[0]: GPIO0 输入 其他配置 Reserved
	0x75	Reserved	7:6	-	R	Reserved
		GPIO_IN MSB	5:0	-	R	GPIO_PDEN LSB[3]: GPIO11 输入 GPIO_PDEN LSB[2]: GPIO10 输入 其他配置 Reserved

12.5 Basic Operation Configuration

page	地址	寄存器	BIT	推荐值	读写	说明
3	0x06	Reserved	7:3	0x00	W/R	Reserved
		REG_TX_CFG_MODE	2		W/R	0: 单次发送模式 1: 连续发送模式
		REG_RX_CFG_MODE	1:0		W/R	0: 单次接收模式 1: 带超时的单次接收模式 2: 连续接收模式
	0x07	REG_RX_TIMEOUT[7:0]	7:0	0x00	W/R	接收超时设置, 低 8 位, 单位微秒
	0x08	REG_RX_TIMEOUT[15:8]	7:0	0x00	W/R	接收超时设置, 高 8 位, 单位微秒
	0x0D	BW	7:4	0x98	W/R	带宽: 6: 62.5 kHz 7: 125 kHz 8: 250 kHz 9: 500 kHz
		CODING RATE	3:1		W/R	码率: 1: 4/5 2: 4/6 3: 4/7 4: 4/8
		Reserved	0		W/R	Reserved
	0x0E	SPREADING FACTOR	7:4	0x78	W/R	SF: 5: 32 chips / symbol 6: 64 chips / symbol 7: 128 chips / symbol 8: 256 chips / symbol 9: 512 chips / symbol 10: 1024 chips / symbol 11: 2048 chips / symbol 12: 4096 chips / symbol
		RXPAYLOADCRC CON	3		W/R	0: CRC 关闭 1: CRC 打开
		Reserved	2:0		W/R	Reserved
	0x0F	SYNCWORD[7:4]	7:5	0x12	W/R	同步字高 4bit

						SF5 时取值范围 0x1~0x 3 SF6 时取值范围 0x 1~0x 7 其余 SF 取值范围 0x 1-0x F
		SYNCWORD[3:0]	3:0			同步字低 4bit SF5 时取值范围 0x0~0x 3 SF6 时取值范围 0x 0~0x 7 其余 SF 取值范围 0x 0-0x F
	0x10	IF[7:0]	7:0	0xC0	W/R	中频配置，次高 8 位
	0x11	IF[11:8]	7:4	0x0F	W/R	中频配置，高 4 位
		Reserved	3:0		W/R	Reserved
	0x12	Reserved	7:4	0x06	W/R	Reserved
		LOWDATARATE	3		W/R	0: 正常模式 1: 低速率模式
		Reserved	2:0		W/R	Reserved
	0x13	PREAMBLE_LEN[7:0]	7:0	0x08	W/R	preamble 长度低 8 位
	0x14	PREAMBLE_LEN[15:8]	7:0	0x00	W/R	preamble 长度高 8 位
	0x1A	SPI3_EN_REG	7	0x03	W/R	3 线 SPI 选择寄存器 0: 不选择 3 线 SPI 模式; 1: 选择 3 线 SPI 模式
		REG_MOSI_PUEN	6		W/R	mosi 管脚的寄存器控制模式 puen 控制寄存器
		REG_CSN_PUEN	5		W/R	csn 管脚的寄存器控制模式 csn 控制寄存器
		REG_SCK_PUEN	4		W/R	sck 管脚的寄存器控制模式 sck 控制寄存器
		REG_IN_PAD_MODE	3		W/R	SPI 输入管脚手动配置模式使能
		REG_OUT_PAD_MODE	2:0		W/R	SPI 的 MISO 管脚模式选择，用于 1 对多的 SPI 应用
	0x24	Reserved	7:4	0x00	W/R	Reserved
		EN_DCDC_3V	3		W/R	DCDC 使能 0: DCDC bypass 1: DCDC 工作
		Reserved	2:0		W/R	Reserved
	0x26	XTAL_ACTIVE_EN_3V	7	0x00	W/R	支持 3V 供电的有源晶振通道使能，保证芯片可工作在晶振 3V 时钟下 1: 打开；0: 关闭
		XTAL_STARTUP_EN_3V	6		W/R	晶振快速启动使能。1: 打开；0: 关闭。
		EN_DIG_COREPOWER_3V	5		W/R	数字核电压使能开关 1: 打开；0: 关闭
		Reserved	4:0		W/R	Reserved

13 Reference Schematic

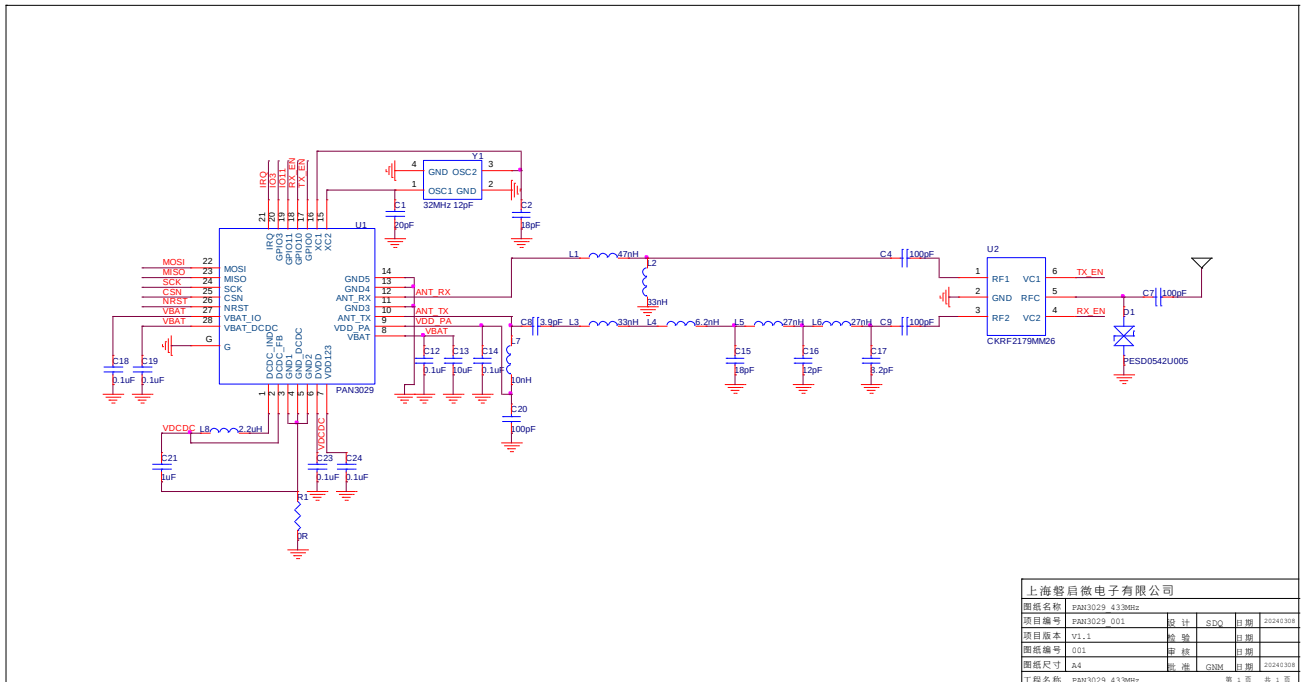


Figure 13-1 433MHz Reference Schematic

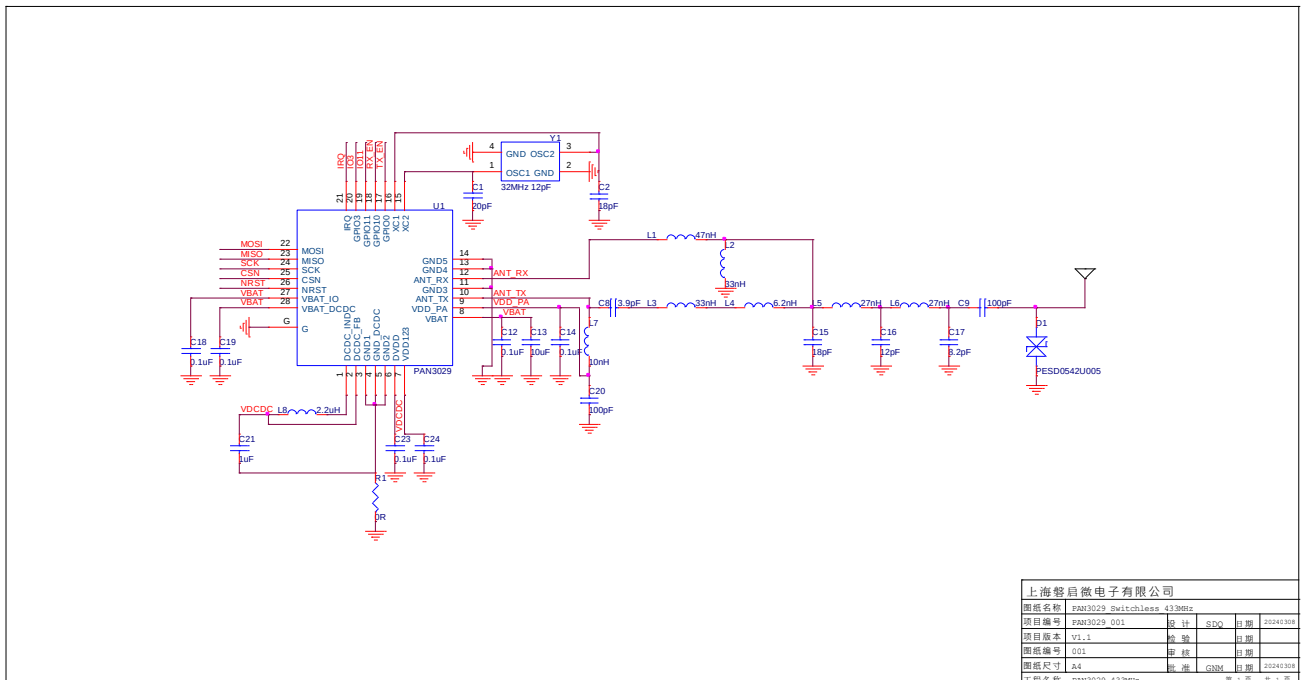


Figure 13-2 Switchless 433MHz Reference Schematic

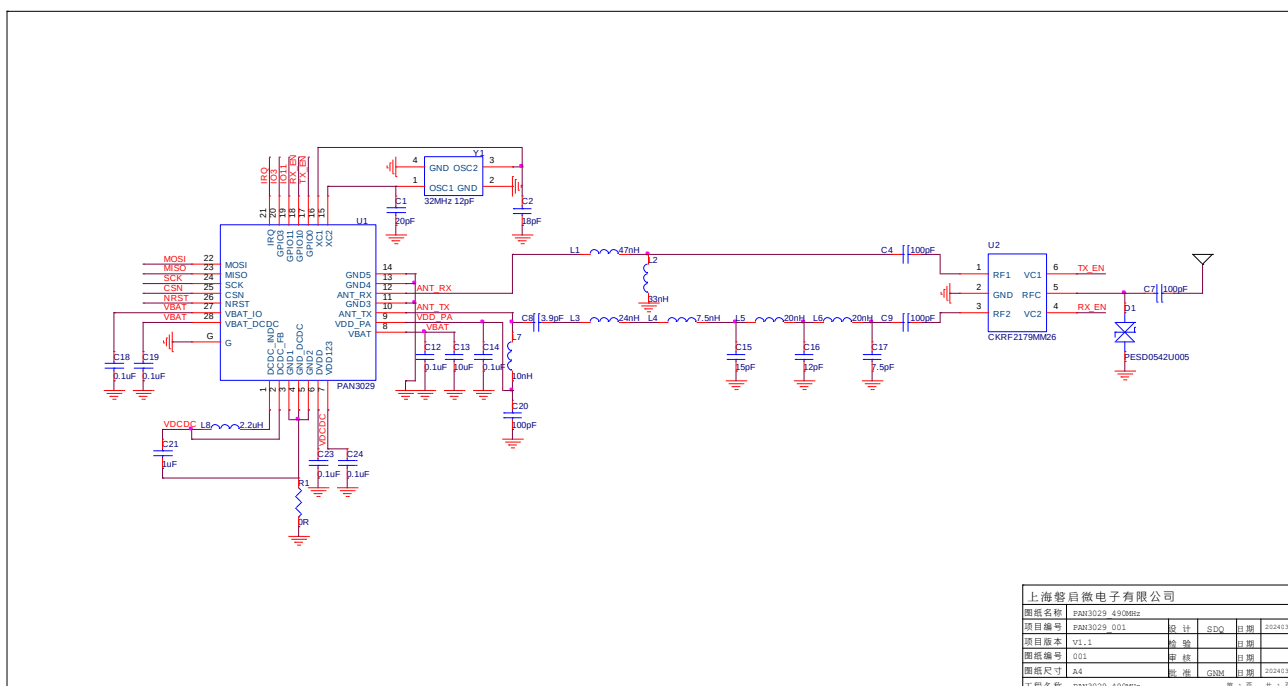


Figure 13-3 490MHz Reference Schematic

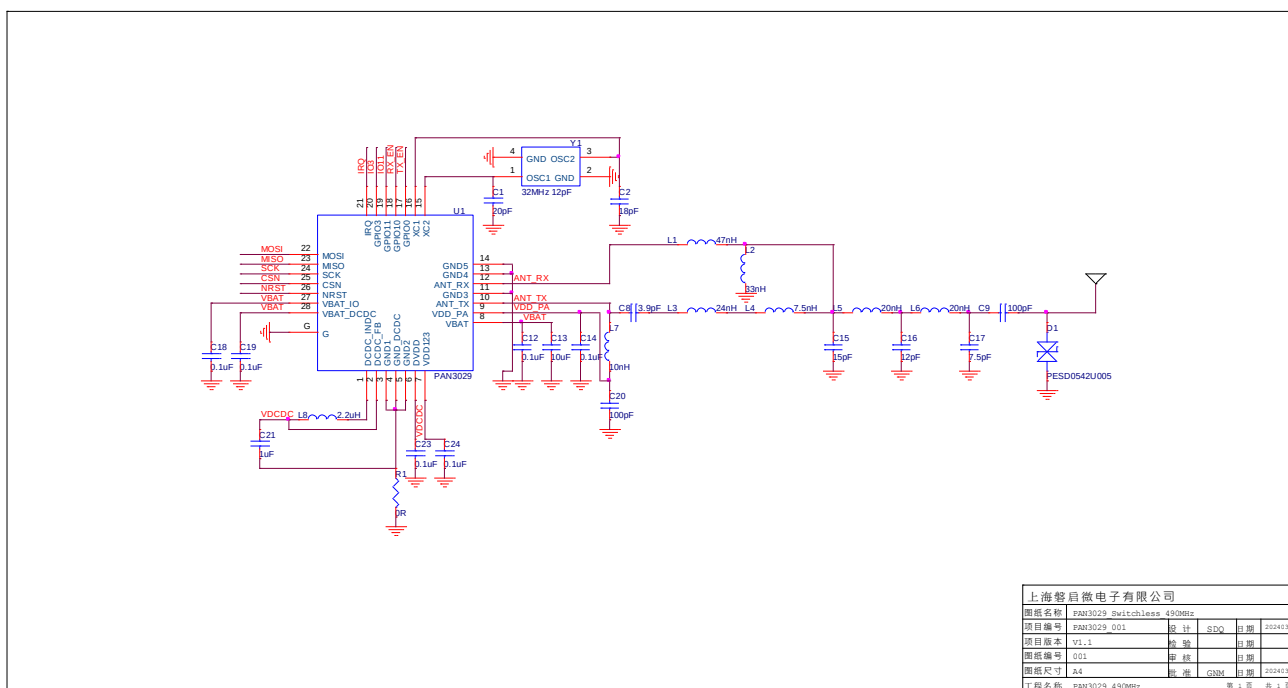


Figure 13-4 Switchless 490MHz Reference Schematic

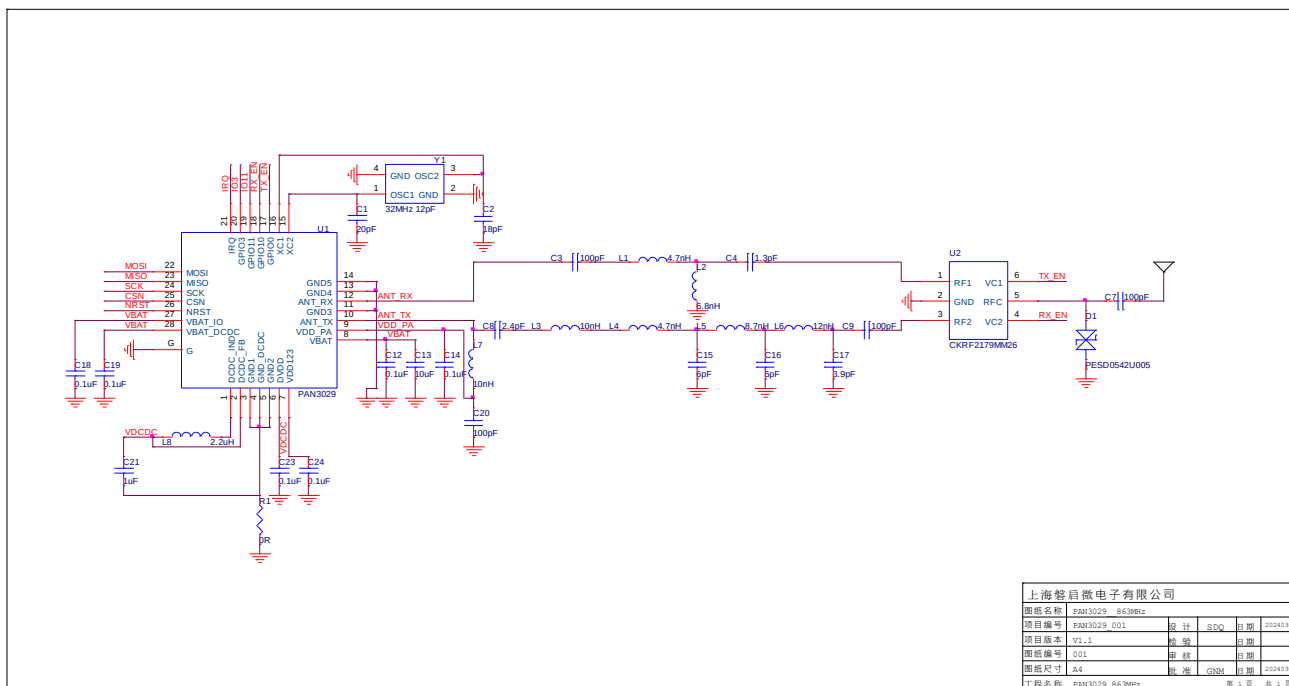


Figure 13-5 863MHz Reference Schematic

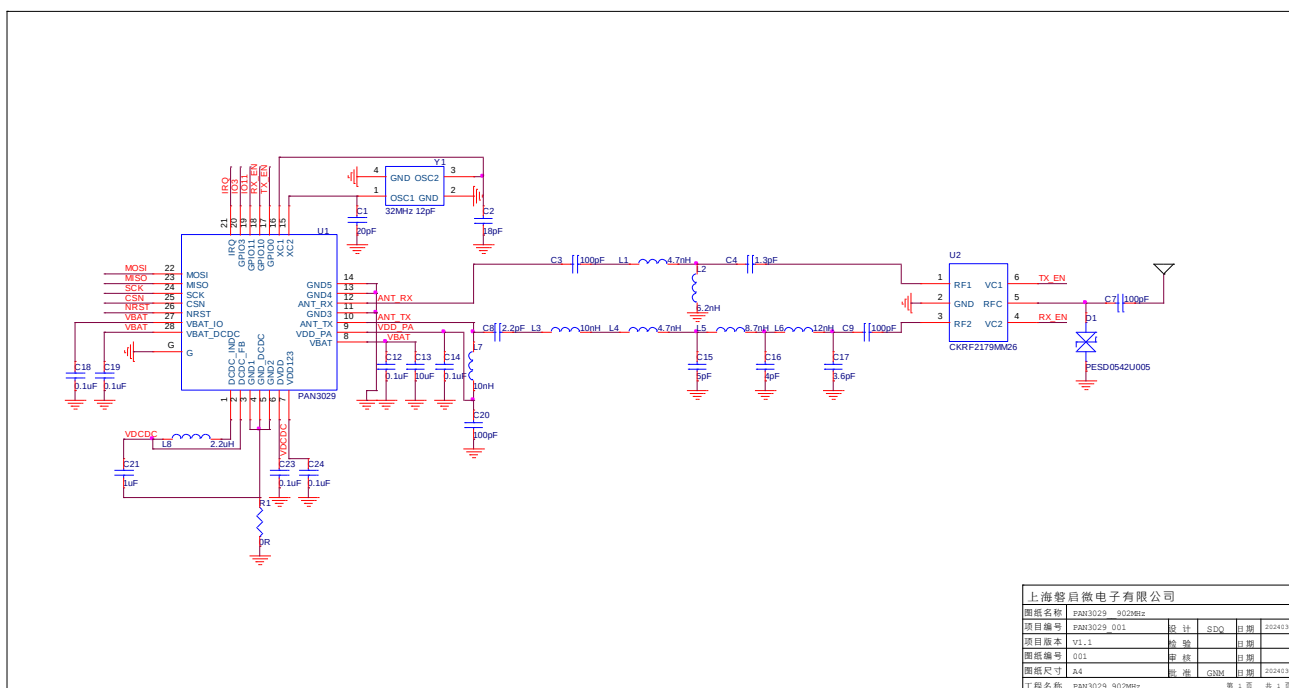


Figure 13-6 915MHz Reference Schematic

14 Package size

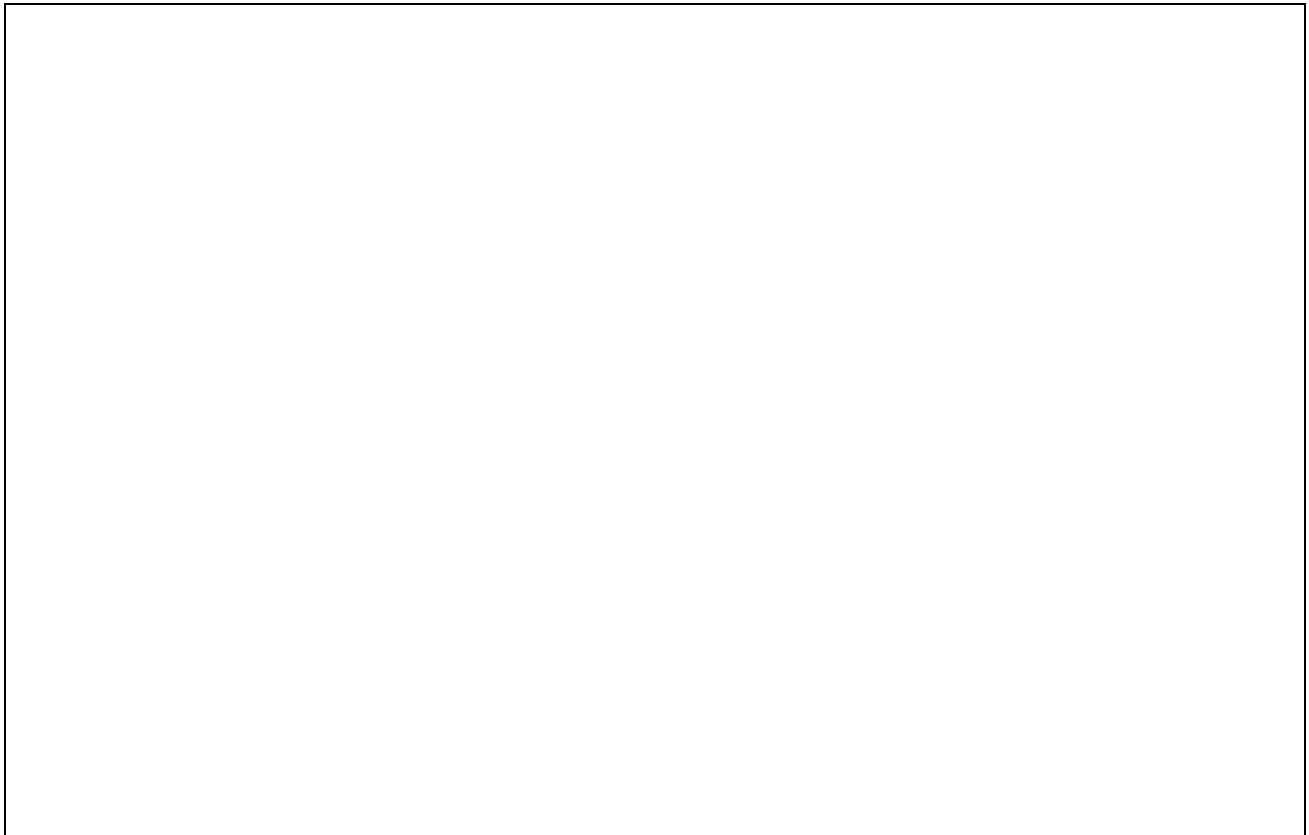


Figure 14-1 QFN28 Package Diagram

Table 14-1 QFN28 package dimensions

符号	最小值(mm)	典型值(mm)	最大值(mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
E	3.90	4.00	4.10
E2	2.55	2.65	2.75
e	0.40 BSC		
K	0.225	0.275	0.325
L	0.35	0.40	0.45
h	0.30	0.35	0.40
Ne	2.40 BSC		
Nd	2.40 BSC		

15 Notes

- 1) 该产品属 CMOS 器件，在储存、运输、使用过程中要注意防静电。
- 2) 器件使用时接地要良好。
- 3) 回流焊温度不能超过 260℃。

无铅回流焊工艺曲线如下图：

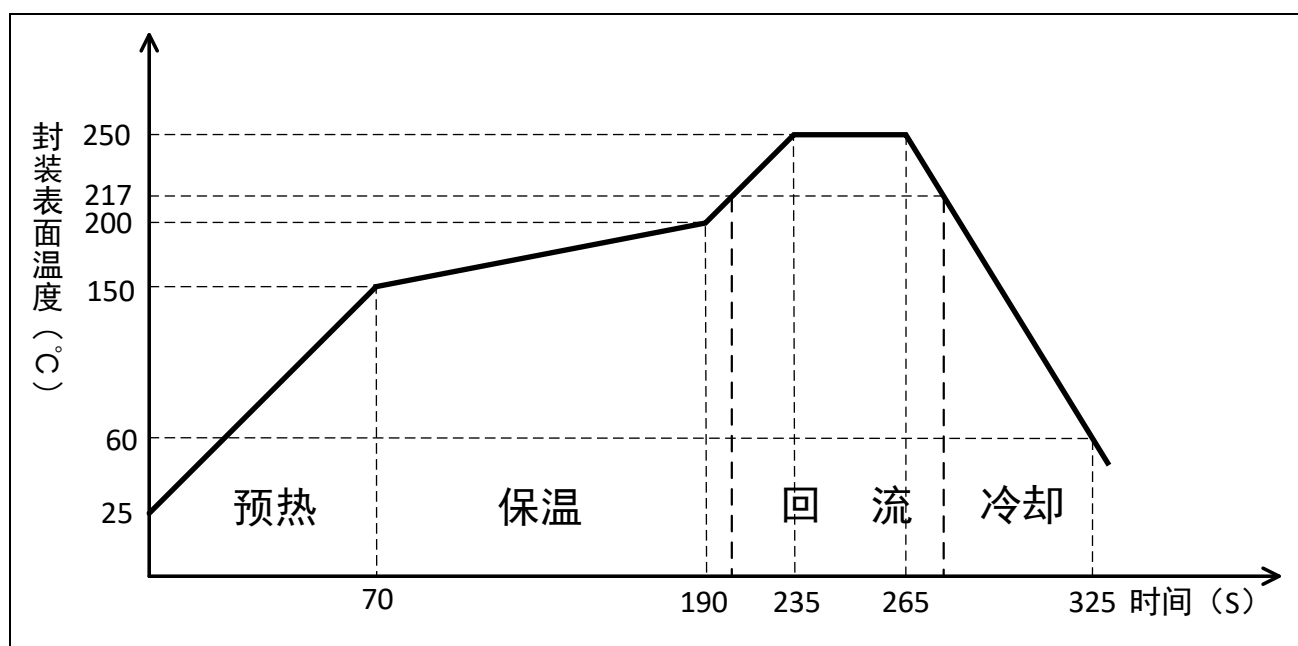


Figure 15-1 Reflow Process Profile

16 Storage conditions

- 1) 产品在密封包装中储存：在温度小于 30°C 且湿度小于 90%时，可达 12 个月。
- 2) 包装袋被打开后，元器件将被回流焊制程或其他的高温制程所采用时必须符合：
 - a) 在 72 小时内且工厂环境为小于 30°C≤60%RH 完成；
 - b) 保存在 10%RH 环境下；
 - c) 使用前进行 125°C，24h 烘烤去除内部水汽。
- 3) MSL（包装湿度敏感性）：3 级（根据 IPC/JEDEC J-STD-020 确定）

Abbreviations

ADC	模数转换器
CAD	信道活跃检测
Chirp	线性调频
CRC	循环冗余校验
CSN	SPI 片选信号
DAC	数模转换器
DCDC	直流变换器
FIFO	先入先出
GPIO	通用型输入输出
IRQ	中断请求
LDO	低压差线性稳压器
LPF	低通滤波器
MAC	介质访问控制层
MCU	微处理单元
Mixer	混频器
Modem	调制解调器
OSC	振荡器
PA	功率放大器
RF	射频
PLL	锁相环
PMU	电源管理单元
POR	上电复位
RAM	随机存取存储器
RSSI	信号强度指示
SCK	SPI 时钟信号
SF	扩频因子
SPI	串行外设接口
STB	待机模式
Sync	同步
VCO	压控振荡器

Revision History

Version	Date	Content
V1.0	2023.05	First edition
V1.1	2023.06	Added packaging method, reference schematic, optimized pin1 and pin2 names in pin definition, and the function remained unchanged
V1.2	2023.07	Added PAN3060 series, electrical characteristics, etc.
V1.3	2023.08	Updated reference schematic, supplemented working voltage description
V1.4	2023.09	Supplemented reference schematics for different frequency bands, supplemented electrical characteristics for different frequency bands
V1.5	2023.10	Updated reference schematic
V1.6	2023.12	Updated the description of SYNCWORD register in basic operation configuration, updated reference schematic
V1.7	2024.03	Updated reference schematic and electrical characteristics.
V1.8	2024.05	Added description of NC in pin description, updated sleep current, transmission current, updated working frequency band
V1.9	2024.08	Increase the operating frequency band

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